Economic Modelling 144 (2025) 107001

Contents lists available at ScienceDirect

Economic Modelling

journal homepage: www.journals.elsevier.com/economic-modelling

Welfare and income effects of tuition subsidies and public investment in schooling☆

Alok Kumar

Department of Economics, University of Victoria, Victoria, British Columbia, Canada, V8W 2Y2

ARTICLE INFO

Dataset link: https://data.mendeley.com/datasets/6t6tjsk66d/1

JEL classification:
C51
C61
E32
E62
H52
O11
O17
O23

Keywords:
Public schooling investment
Tuition subsidies
Expenditure multipliers
Welfare
Poor households
India

ABSTRACT

Evidence demonstrates that poorer individuals have lower human capital attainment than richer individuals. This study constructs and estimates a two-sector dynamic stochastic general equilibrium model with human capital to analyse the welfare and income effects of different types of public schooling expenditure in India. The results show that tuition subsidies that are targeted towards the poor have a significantly positive effect on households' income and welfare. Although the rich suffer welfare loss, aggregate welfare rises. Public schooling investment and untargeted tuition subsidies have a significantly positive effect on poor households' income and welfare; however, the effect is smaller relative to tuition subsidies targeted at the poor. Public schooling investment has a larger impact on poor households' human capital and income.

1. Introduction

Low human capital is a major impediment to economic development and poverty reduction. Most developing countries are characterised by three important and interrelated features of low human capital attainment, high incidence of poverty and a large informal sector in economic activity. Empirical literature demonstrates that poorer households have lower human capital attainment than richer households (UNESCO, 2008). For example, UNESCO-WIDE (2022) data shows that the mean year of schooling for households in the first (lowest) quintile of wealth was 4.20 years, while it was 9.86 years for households in the fifth (highest) quintile in low income countries for the 2003–2015 period (Table 1).[1] Similarly, the school completion rate at the primary level for children from households in the first quintile was only 32%, while it was 80% for children from households in the fifth quintile.

Empirical literature also shows that children from poorer households have inferior learning outcomes compared with children from more affluent households enrolled in the same grade (Orazem and King, 2007; UNESCO, 2008). This is partially attributable to families' inability to afford higher quality schools or families living in neighbourhoods with schools of lower quality. Therefore, poorer children may have no choice but to attend schools with inferior learning environments. The other significant barrier faced by poor households is inadequate access to credit markets. A large body of theoretical and empirical literature shows that a lack of access to credit markets is one of the most significant causes of inferior schooling and high incidence of child labour among poor families (Baland and Robinson, 2000; Orazem and King, 2007; Jacoby and Skoufias, 1997; Kumar, 2013).

Poor and low income earners are predominantly employed in the informal sector, which employs 70% of the labour force in a typical developing country and accounts for one-third of GDP (Loayza, 2016; World Bank, 2022). This sector is predominantly engaged in producing non-traded goods and services using labour-intensive technologies and

☆ I thank the editor and two anonymous referees for their numerous incisive comments that have significantly improved the paper. The responsibility for any errors remaining is entirely mine.

E-mail address: kumara@uvic.ca.

[1] UNESCO-WIDE data are available at https://www.education-inequalities.org/.

https://doi.org/10.1016/j.econmod.2024.107001
Received 28 February 2024; Received in revised form 30 December 2024; Accepted 31 December 2024
Available online 7 January 2025
0264-9993/© 2025 The Author. Published by Elsevier B.V. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

Table 1
Educational attainment by quintile groups.
Source: UNESCO-WIDE Data.

Low income countries (2003–2015)				
Quintile	Primary Comp.	Lower-Sec Comp.	Upper-Sec. Comp.	Edu Years
1st	0.32	0.14	0.09	4.20
2nd	0.40	0.19	0.10	5.12
3rd	0.49	0.22	0.14	5.75
4th	0.63	0.35	0.21	7.07
5th	0.80	0.61	0.48	9.86
Lower-Middle-Income Countries (2003–2015)				
Quintile	Primary Comp.	Lower-Sec Comp.	Upper-Sec. Comp.	Edu Years
1st	0.66	0.46	0.13	5.19
2nd	0.80	0.62	0.24	7.40
3rd	0.88	0.74	0.36	9.00
4th	0.93	0.82	0.50	10.53
5th	0.96	0.91	0.72	12.66
Upper Middle Income Countries (2003–2015)				
Quintile	Primary Comp.	Lower-Sec Comp.	Upper-Sec. Comp.	Edu Years
1st	0.91	0.79	0.48	9.05
2nd	0.96	0.86	0.57	10.54
3rd	0.96	0.87	0.62	10.88
4th	0.96	0.90	0.62	11.52
5th	0.98	0.94	0.72	12.37
India (2015)				
Quintile	Primary Comp.	Lower-Sec Comp.	Upper-Sec. Comp.	Edu Years
1st	0.80	0.59	0.13	5.18
2nd	0.91	0.75	0.25	7.50
3rd	0.94	0.84	0.38	9.17
4th	0.97	0.90	0.53	10.75
5th	0.98	0.96	0.76	12.93

Note: Primary Comp = Completed Grade V, Lower-Sec Comp. = Completed Grade VIII and Upper-Sec. Comp. = Completed Grade XII.

employing lower-skilled and lower-paid workers (La Porta and Shleifer, 2014; ILO, 2018; World Bank, 2022). In addition, the rate of return on human capital in this sector is lower compared with the formal sector (Gong and van Soest, 2002; Kuepie et al., 2009; Unni, 2023).

This study analyses the income and welfare effects of public expenditure on human capital/schooling and whether it has a differential effect on poor and rich households' schooling investment. Public expenditure on education can take many forms such as direct financial support to schools and subsidising schooling/tuition costs. Tuition subsidies can be general and made available to rich and poor households, but they can also be targeted to a particular type of households. In addition, considering credit constraints, income support for poor households can also encourage schooling investment. In addition, this study examines the income and welfare effects of different types of public education expenditure.

This study constructs and estimates a closed economy dynamic stochastic general equilibrium (DSGE) model with human capital. The model includes formal and informal sectors. The estimations employ the Bayesian full-information maximum likelihood method. In the model, the production of human capital depends on households' time spent on schooling and public investment in schooling. Households are either rich or poor. The schooling system consists of high quality/productivity and low quality/productivity schools. Poor households do not participate in financial markets and only have access to lower quality schools. Apart from public investment in schools, the government provides tuition subsidies to all households and income support for poor households, financing expenditures using a mix of lump-sum and distortionary taxes and borrowing. I estimate the model using annual macro-aggregate and school enrolment data for India from 1992–2019.

The estimated model matches well the dynamics of macro-aggregates and school enrolment in India. The estimation results show that data favour models with lower elasticity of human capital to households' time spent schooling (≤ 0.20) over models with higher elasticity. In addition, the elasticity of human capital to government investment in schooling is larger for poor households compared with rich households, indicating that government investment in schooling has a larger impact on poor households' human capital in India.

The results show that among all the policies analysed, tuition subsidies targeted at poor households have the largest positive effect on the income and welfare with a long-term mean income multiplier exceeding one. Although public schooling investment and untargeted tuition subsidies have a significantly positive effect on poor households' income and welfare, their effect is smaller relative to tuition subsidies targeting the poor. Poor households' long-term income multipliers with respect to public schooling investment and untargeted tuition subsidies are estimated to be approximately one and 0.44, respectively. Poor households' welfare gain is approximately 25%, relative to tuition subsidies targeting the poor.

The lump-sum income transfer to poor increases poor households' human capital, consumption and welfare but has a negative effect on employment income with an employment income multiplier that is negative across all horizons. The results also reveal that all the spending policies analysed in this study, with the exception of tuition subsidies targeted at rich households have a negative effect on welfare. Despite this welfare loss, aggregate welfare rises significantly in the case of tuition subsidies targeted at poor households and untargeted tuition subsidies. The estimates show that between these two policies, the aggregate welfare gain from the targeted tuition subsidy is about three and half times more than that from the untargeted tuition subsidy.

To the best of my knowledge, this is the first study to analyse the welfare and income effects of different types of public expenditure on schooling using a two-sector DSGE model with heterogeneous households and schooling quality. This study relates to research that analyses the cyclicality of schooling using the Real Business Cycle framework (Perli and Sakellaris, 1998; DeJong and Ingram, 2001). Perli and Sakellaris (1998) investigate the contribution of schooling investment in accounting for GDP persistence in the United States (US).

DeJong and Ingram (2001) analyse the effects of productivity shocks to goods and human capital sectors on private schooling investment in the US. These studies neglect households' and schools' heterogeneity and do not examine effects of public expenditure on schooling.

The remainder of this paper is organised as follows. Section 2 presents the model. Section 3 details the model's calibration process and the empirical estimation approach. Section 4 presents the results from the posterior estimation and examines the model's performance. Section 5 analyses the effects of various public policies and presents estimates of multipliers and welfare costs/benefits using posterior distributions. This is followed by the conclusion. For brevity, I only present the main equations in the text, and all equilibrium equations are listed in Appendix.

2. Model

The model is a closed economy DSGE model with human capital, formal (f) and informal (n) sectors and rich (R) and poor (P) households. In the model, human capital affects output as a factor of production as in the growth accounting and human capital augmented Solow model.[2] Fraction $\omega \in (0,1)$ of households are rich and $1-\omega$ are poor. Rich households own all the firms operating in formal and informal sectors. Both types of households consume formal and informal goods, supply labour and acquire human capital. Human capital is assumed to be embodied in labour, similar to Perli and Sakellaris (1998) and DeJong and Ingram (2001).

Human capital production depends on time devoted by households on schooling and government investment in schooling. Households also incur financial cost to acquire human capital. In addition to investment in schooling, the government also consumes and provides lump-sum transfers to poor households.[3] The government finances expenditures through taxes and borrowing. Consistent with the evidence, taxes are imposed only on formal sector goods and incomes.

Poor households differ from rich households in three key respects. First, the discount factor of rich households (β_R) is higher than that of poor households (β_P). This assumption is consistent with previous empirical evidence demonstrating a positive relationship between wealth/income and patience (Meier and Sprenger, 2010; Tanaka et al., 2010; Warner and Pleeter, 2001). With the assumption that rich households are more patient, in equilibrium, such households would prefer to save and poor households would prefer to borrow. However, imperfect financial markets prevent poor households from borrowing. This allows me to capture the effects of borrowing constraints on poor households' human capital investment. As noted previously, theoretical and empirical literature shows that a lack of access to financial markets can adversely affect household choices of schooling.

Second, poor households have access to lower quality schools compared with rich households. To capture this differing access, I assume that the human capital sector consists of two sub-sectors: R and P. Schools in sub-sector R have higher human capital productivity and are more costly than schools in sub-sector P. Rich (poor) households attend higher (lower) productivity schools. Subsequently, I refer to low-quality schools as poor schools and high-quality school as rich schools.

Finally, rich households can work in the formal sector as employees or as entrepreneurs/self-employed workers in the informal sector; however, poor households can only work as employees in the informal sector. This is consistent with the evidence that poorer households have low wage earnings, are primarily engaged in low productivity activities and the average skill level of workers in informal sector is lower.

Let s_{jt} and g_{ht} be time spent schooling by $j = R$ & P type of households and government schooling investment at time t, respectively. Then, the laws of human capital motion for representative rich (h_{Rt}) and poor (h_{Pt}) households are obtained as follows:

$$h_{Rt+1} = (1-\delta_h)h_{Rt} + A_{Rht}{s_{Rt}}^{\theta}{g_{ht}}^{\theta_{gR}} \ \& \tag{2.1}$$

$$h_{Pt+1} = (1-\delta_h)h_{Pt} + \kappa_h A_{Pht}{s_{Pt}}^{\theta}{g_{ht}}^{\theta_{gP}} \tag{2.2}$$

where $\theta > 0$ and $\theta_{gj} > 0$ are the elasticity of human capital production with respect to households' time spent schooling and the elasticity of human capital production with respect to government schooling investment in $j = R$ & P type of schools, respectively. $\kappa_h \in (0,1)$ is the productivity parameter of Pth type of human capital sector, which captures lower productivity. A_{jht} and δ_h are exogenous stochastic productivity processes in the jth human capital sector and human capital depreciation rate, respectively. Due to lack of data on human capital obsolescence, I assume an identical depreciation rate for each type of households.[4]

Goods in the formal sector are produced by firms (unit measure) using capital (k_{ft}) and effective labour (el_{ft}). Formal sector firms hire workers in the formal labour market. The formal sector production function (y_{ft}) is as follows:

$$y_{ft} = A_t k_{ft}^{\alpha_f} el_{ft}^{1-\alpha_f} \text{ with } 0 < \alpha_f < 1 \tag{2.3}$$

where A_t is the exogenous stochastic total factor productivity (TFP) process in the formal sector. Formal goods are used for private consumption, investment and schooling and public expenditure.

Informal goods are produced by self-employed workers from rich households using their own effective labour (el_{Rnt}) and poor households' effective labour (el_{Pnt}) working in the informal labour market. The informal sector production function (y_{nt}) is as follows:

$$y_{nt} = \kappa_n A_t {el_{Rnt}}^{\alpha_n} {el_{Pnt}}^{1-\alpha_n} \text{ with } 0 < \alpha_n \ \& \ \kappa_n < 1 \tag{2.4}$$

where κ_n is the productivity parameter, and (2.4) captures the low productivity, low-skill intensity and highly labour-intensive nature of

[2] Broadly speaking, three alternative frameworks have been used to analyse the influence of human capital on economic growth: (i) human capital as a factor of production as in the growth accounting and the augmented Solow model, (ii) human capital as a facilitator for the adoption and diffusion of new technologies (Nelson and Phelps, 1966; Barro, 1991; Benhabib and Spiegel, 1994) and (iii) human capital based endogenous growth models in which human capital is needed for new innovations and the introduction of new goods (Romer, 1986, 1990) or it can have spillover effects (Lucas, 1988).

In general, the empirical evidence suggests that the endogenous growth model is more likely to be appropriate for advanced countries, where the majority of research and development activities are concentrated. For developing countries such as India, the first two approaches are more likely to be applicable (Benhabib and Spiegel, 1994; Mankiw et al., 1992; Rossi, 2020; Sianesi and Reenen, 2003).

[3] For expositional purposes, I analyse the issue of tuition subsidies in Section 5.

[4] Human capital obsolescence can arise for many reasons (see De Grip and van Loo (2002) and De Grip (2006) for a thorough review). It can be attributable to (i) natural ageing process, retirements, injuries and illness (wear of skills); (ii) insufficient use of skills, i.e. atrophy (e.g. mismatch between skills and employment, over-qualification in which an employee is working below their attained education and unemployment); (iii) job-specific skill obsolescence related to changes in technological or organisational production processes; (iv) skill obsolescence due to sectoral shifts in employment; and (v) firm-specific skill obsolescence, which may occur due to firms' closure or reorganisation.

The relative significance of sources of human capital obsolescence is likely to differ across sectors. The informal sector is highly labour-intensive and has a low capital and technology base, with most workers engaged in manual work. In this sector, technical obsolescence caused by atrophy (particularly for rich households) and wear due to physically demanding jobs, illness and injury (particularly for poor households) is more likely to occur. Economic obsolescence, particularly job-specific skill obsolescence, is more likely to occur in the formal sector, which is capital-intensive, employs advanced technologies and is more exposed to information and communications technology changes. I thank an anonymous referee for suggesting that I include a discussion concerning the sources of human capital obsolescence.

the informal sector in a simple way. Informal goods are only used for domestic private consumption.

Let l_{Rit} be rich households' labour supply in production sector $i = f$ & n. Then the aggregate effective labour supply to the formal sector (el_{ft}) and to the informal sector by poor households (el_{Pnt}) and rich households (el_{Rnt}) are as follows:

$$el_{ft} = \omega h_{Rt} l_{Rft}; \tag{2.5}$$

$$\omega el_{Pnt} = (1-\omega) h_{Pt} l_{Pt} \ \& \tag{2.6}$$

$$el_{Rnt} = h_{Rt} l_{Rnt}. \tag{2.7}$$

where l_{Pt} is the labour supplied by poor households to the informal sector.[5]

Households

Households have the following logarithmic preferences:

$$\ln c_{jt} - \mu_j \frac{(l_{jt} + \phi_{js} s_{jt})^{1+\zeta}}{1+\zeta} \text{for } j = R, P \tag{2.8}$$

where c_{jt}, l_{jt} and μ_j denote the private consumption index, the labour supply index and the relative weight of time spent working and schooling in the utility function of jth type of households, respectively. ϕ_{js} captures the disutility incurred in schooling relative to working.[6] ζ is the inverse of elasticity of time spent working and schooling with respect to wages.

The private consumption index is the constant elasticity of substitution aggregator of formal and informal goods consumption (c_{jit} for $i = f$ & n), which is as follows:

$$c_{jt} = \left[\chi^{1/\tau} c_{jft}{}^{\tau-1/\tau} + (1-\chi)^{1/\tau} c_{jnt}{}^{\tau-1/\tau}\right]^{\tau/\tau-1}$$
$$\text{with } 0 < \chi < 1 \ \forall \ j = R \ \& \ P \tag{2.9}$$

where c_{jft} and c_{jnt} respectively represent formal and informal goods consumption by the jth type of household at time t. χ and τ are the weight of formal goods in the consumption index and the elasticity of substitution between formal and informal goods in consumption, respectively.

The price of a composite consumption basket (p_t) is used as the numeraire. Let p_{ft} and p_{nt} be the (relative) respective prices of formal and informal goods. Then, the optimal intra-temporal allocation implies the following:

$$1 = \chi((1+tc)p_{ft})^{1-\tau} + (1-\chi)p_{nt}^{1-\tau} \tag{2.10}$$

where tc is the sales tax on formal goods.

The demand functions of formal and informal goods by the jth type of households are as follows:

$$c_{jft} = \chi c_{jt}((1+tc)p_{ft})^{-\tau} \ \& \tag{2.11}$$

$$c_{jnt} = (1-\chi) c_{jt} p_{nt}^{-\tau} \tag{2.12}$$

for $j = R$ & P.

Rich households

Rich households supply labour to formal and informal sectors, physical capital to the formal sector and make human capital and financial investments in terms of a one-period government bond. The labour supply index for a rich household is defined as follows:

$$l_{Rt} = \phi_{lf} l_{Rft} + l_{Rnt} \tag{2.13}$$

where ϕ_{lf} denotes the relative disutility of working in the formal sector, which helps in matching the wage differential between formal and informal sectors observed in India.

The representative rich household enters time period t with units of government bond holding (b_t), physical capital stock (k_{Rt}) and human capital stock (h_{Rt}). At the beginning of period t, the rich household chooses its consumption (c_{Rt}), time spent working in the formal (l_{Rft}) and informal (l_{Rnt}) sectors, time spent schooling (s_{Rt}), the amount of informal effective labour to hire (el_{Pnt}), the amount of government bond holding in the next period (b_{t+1}) and the amount of physical capital investment (I_t). While making these decisions, the rich household considers the prices of formal and informal goods, the wage per unit of effective labour in the formal sector (w_{ft}), the wage per unit of effective labour in the informal sector (w_{nt}), the per-unit cost of schooling as a formal good (v_{Rt}), the rental rate on physical capital (rk_{ft}) and the (gross) interest rate (r_{t+1}) on the government bond between periods t and $t+1$.

The representative rich household maximises its expected utility:

$$E_0 \sum_{t=0}^{\infty} \beta_R^t \gamma_t \left[\ln c_{Rt} - \mu_R \frac{(l_{Rt} + \phi_{Rs} s_{Rt})^{1+\zeta}}{1+\zeta}\right]$$

subject to the following budget constraint:

$$(1-tw) w_{ft} h_{Rt} l_{Rft} + [p_{nt} y_{nt} - el_{Pnt} w_{nt}] + r_t b_t + (1-tk) rk_{ft} k_{Rft} + \frac{div_t}{\omega} =$$

$$c_{Rt} + p_{ft} v_{Rt} s_{Rt} + (1+tc) p_{ft} I_t + b_{t+1} + to_t \ \forall \ t \geq 0; \tag{2.14}$$

the law of motion for physical capital:

$$k_{Rt+1} = (1-\delta_k) k_{Rt} + I_t; \tag{2.15}$$

(2.1), (2.9), (2.11), (2.12), (2.13), (2.14) and (2.15) given prices, taxes and laws of motion for exogenous stochastic processes. γ_t, to_t and div_t are the preference shock, lump-sum tax and total dividend paid by formal firms at time t, respectively. tw, tk and δ_k are tax rates imposed on formal sector wage income and capital income and the depreciation rate of capital, respectively.[7]

Let λ_{ct}^R and λ_{ht}^R be the Lagrange multipliers associated with the budget constraint (2.14) and the law of motion for human capital of rich household (2.1), respectively. Then from the first-order conditions for l_{Rft}, l_{Rnt} and s_{Rt} we obtain the following:

$$(1-tw) w_{ft} = \phi_{lf} MPEL_{Rt} \tag{2.16}$$

where $MPEL_{Rt} \equiv \alpha_n p_{nt} \frac{y_{nt}}{el_{Rnt}}$ denotes the marginal product of the rich household' s effective labour of in the informal sector to obtain the following:

$$(p_{ft} v_{Rt} + \phi_{Rs} MPEL_{Rt} h_{Rt}) \lambda_{ct}^R = \theta A_{hRt} s_{Rt}{}^{\theta-1} g_{ht}{}^{\theta_{gR}} \lambda_{ht}^R. \tag{2.17}$$

where (2.16) equalises the (net) rate of return from rich households supplying labour to formal and informal sectors. (2.17) represents rich households' trade-off between working and schooling. The left hand side of (2.17) is the opportunity cost of schooling that includes financial cost and net loss in income. The right hand side is the gain in terms of value of human capital.

The shadow value of human capital for rich households is obtained as follows:

$$\lambda_{ht}^R = \beta_R E_t[((1-tw) w_{ft+1} l_{Rft+1} + MPEL_{Rt+1} l_{Rnt+1}) \lambda_{ct+1}^R + (1-\delta_h) \lambda_{ht+1}^R]. \tag{2.18}$$

[5] Each rich household is assumed to own and operate one informal business; thus, the measure of informal businesses is equal to ω.

[6] The inclusion of time spent in school in the utility function captures the trade-off between working and schooling (DeJong and Ingram, 2001; Perli and Sakellaris, 1998). Due to prevalence of child labour, this trade-off is likely to occur at a much earlier age in poorer countries. Poverty and credit constraints are some of the most significant factors identified in the empirical literature as being correlated with a higher incidence of child labour and lower schooling participation and attainment.

[7] For simplicity, I assume fixed distortionary tax rates. In response to any-shock, lump-sum tax/transfer (to_t) and government borrowing (b_{t+1}) adjust to maintain a balanced government budget.

Poor households' effective labour demand for informal production is obtained as follows:

$$w_{nt} = (1-\alpha_n)p_{nt}\frac{y_{nt}}{el_{Pnt}}. \tag{2.19}$$

For future use, I define the rich household's net employment income (inclusive of income from self-employment) (ein_{Rt}) as follows:

$$ein_{Rt} = (1-tw)w_{ft}h_{Rt}l_{Rft} + [p_{nt}y_{nt} - el_{Pnt}w_{nt}]. \tag{2.20}$$

Poor households

The representative poor household enters time period, t, with its human capital stock, h_{Pt}. At the beginning of period t, the poor household chooses its consumption (c_{Pt}), time spent working (l_{Pt}) and time spent schooling (s_{Pt}). The representative poor household maximises its expected utility:

$$E_0\sum_{t=0}^{\infty}\beta_P^t\gamma_t\left[\ln c_{Pt} - \mu_P\frac{(l_{Pt}+\phi_{Ps}s_{Pt})^{1+\zeta}}{1+\zeta}\right]$$

subject to the following budget constraint:

$$w_{nt}h_{Pt}l_{Pt} + p_{ft}g_{Pt} = c_{Pt} + p_{ft}v_{Pt}s_{pt}\ \forall\ t \geq 0 \tag{2.21}$$

(2.2), (2.11) and (2.12) given prices, taxes/transfers and laws of motion for exogenous stochastic processes. g_{Pt} is the government' s lump-sum transfer to poor households at time t and v_{Pt} is the per-unit schooling cost in terms of formal goods for the poor household at time t.

From the first-order conditions of l_{pt} and s_{Pt}, we obtain the following:

$$(p_{ft}v_{Pt} + \phi_{Ps}w_{nt}h_{Pt})\lambda_{ct}^P = \kappa_h\theta A_{hPt}{s_{Pt}}^{\theta-1}{g_{ht}}^{\theta_{gp}}\lambda_{ht}^P \tag{2.22}$$

where λ_{ct}^P and λ_{ht}^P are the Lagrange multipliers associated with the poor household' s budget constraint (2.21) and the law of motion for human capital of (2.2), respectively. (2.22) shows the poor household' s trade-off between working and schooling.

The shadow value of human capital for poor households is obtained as follows:

$$\lambda_{ht}^P = \beta_P E_t[w_{nt+1}l_{Pt+1}\lambda_{ct+1}^P + (1-\delta_h)\lambda_{ht+1}^P]. \tag{2.23}$$

Formal sector production

The profit function of the representative firm at time t is:

$$p_{ft}y_{ft} - w_{ft}el_{ft} - \frac{\psi}{2}\left(\frac{el_{ft}}{el_f}-1\right)^2 el_f - rk_{ft}k_{ft}$$

where firms incur quadratic adjustment cost in adjusting labour. The adjustment cost captures the cost associated with labour market regulations faced by formal sector firms in a simple way. e_{lf} is the steady-state value of effective labour in the formal sector. Throughout the study variables without subscript t indicate the steady-state value.

The optimal choices of formal firms are characterised as follows:

$$w_{ft} + \psi\left(\frac{el_{ft}}{el_f}-1\right) = (1-\alpha_f)p_{ft}\frac{y_{ft}}{el_{ft}}\ \& \tag{2.24}$$

$$rk_{ft} = \alpha_f p_{ft}\frac{y_{ft}}{k_{ft}}. \tag{2.25}$$

Government

The government budget constraint at time t is:

$$p_{ft}g_{ht} + p_{ft}g_{ct} + (1-\omega)p_{ft}g_{Pt} + r_t b_t = tcp_{ft}(\omega c_{Rft} + (1-\omega)c_{Pft}) + \omega(twl_{Rft}h_{Rt}w_{ft} + tkrk_{ft}k_{Rt} + to_t) + b_{t+1} \tag{2.26}$$

where g_{ct} is the government consumption expenditure at time t.

The stochastic processes of government schooling investment, lump-sum transfer to poor households and government consumption expenditure are respectively as follows:

$$\ln\left(\frac{g_{ht}}{g_h}\right) = \rho_{gh}\ln\left(\frac{g_{ht-1}}{g_h}\right) + e_{ght}; \tag{2.27}$$

$$\ln\left(\frac{g_{pt}}{g_P}\right) = \rho_{gP}\ln\left(\frac{g_{Pt-1}}{g_P}\right) + e_{gPt}\ \& \tag{2.28}$$

$$\ln\left(\frac{g_{ct}}{g_c}\right) = \rho_{gc}\ln\left(\frac{g_{ct-1}}{g_c}\right) + e_{gct}. \tag{2.29}$$

Other stochastic processes

$$\ln A_t = \rho_A\ln A_{t-1} + e_{At}; \tag{2.30}$$

$$\ln\left(\frac{A_{jht}}{A_{jh}}\right) = \rho_{Ah}\ln\left(\frac{A_{jht-1}}{A_{jh}}\right) + e_{jAht}\ \forall j = R\ \&\ P; \tag{2.31}$$

$$\ln\left(\frac{v_{jt}}{v_j}\right) = \rho_{vj}\ln\left(\frac{v_{jt-1}}{v_j}\right) + e_{jvt}\ \forall j = R\ \&\ P\ \& \tag{2.32}$$

$$\gamma_t = \rho_\gamma\gamma_{t-1} + e_{\gamma_t}. \tag{2.33}$$

Aggregation and equilibrium

This study uses a standard definition of equilibrium. The economy satisfies the following market clearing conditions and resource constraints:

$$y_{nt} = \omega c_{Rnt} + (1-\omega)c_{Pnt}. \tag{2.34}$$

$$y_{ft} = \omega c_{Rft} + (1-\omega)c_{Pft} + \omega I_t + g_{ht} + g_{ct} + \omega v_{Rt}s_{Rt} + (1-\omega)v_{Pt}s_{Pt}. \tag{2.35}$$

Aggregate demand for capital stock equals its supply as follows:

$$\omega k_{Rft} = k_{ft}\ \& \tag{2.36}$$

$$gdp_t = p_{nt}y_{nt} + p_{ft}y_{ft} - \omega p_{ft}v_{Rt}s_{Rt} - (1-\omega)p_{ft}v_{Pt}s_{Pt} - \frac{\psi}{2}\left(\frac{el_{ft}}{el_f}-1\right)^2 el_f. \tag{2.37}$$

Aggregate private consumption expenditure (pc_t) is obtained as follows:

$$pc_t = p_{ft}(\omega c_{Rft} + (1-\omega)c_{Pft}) + p_{nt}y_{nt}. \tag{2.38}$$

3. Empirical approach

I use a Bayesian framework to estimate models employing the following six observables: (i) real GDP, (ii) real private consumption expenditure, (iii) real private physical investment expenditure, (iv) real government expenditure on schooling, (v) school enrolment rate in classes I–VIII and (vi) school enrolment rate in tertiary education from 1992 to 2019 in India.[8] I use post-1992 data because major structural reforms occurred in India in 1991 aimed at de-licensing and deregulating the Indian economy. Evidence suggests that these reforms had significant effect on the dynamics of the Indian economy (Ghate et al., 2013; Kumar, 2023).

All data series except for the real government expenditure on schooling, school enrolment rate in classes I–VIII and school enrolment rate in tertiary education are obtained from the *Database of Indian*

[8] Tertiary education refers to post-grade XII education, encompassing diploma courses and undergraduate and graduate studies.

Table 2
Calibrated values of parameters that are common across models.

Parameters	Value
Rich Households' Discount Rate , β_R	0.96
Poor Households' Discount Rate , β_P	0.95
Parameter of Share of Formal Goods in Consumption, χ	0.20
Inverse Elasticity of Time spent Working and Schooling, ζ	2.00
Elasticity of Physical Capital in Formal Production, α_f	0.33
Elasticity of Rich Households' Effective Labor in Informal Production, α_n	0.2024
Physical Capital Depreciation Rate, δ_k	0.10
Human Capital Depreciation Rate, δ_h	0.01
Relative Disutility of Working in the Formal Sector, ϕ_{lf},	6.02527
Tax Rate on Wage Earnings, tw	0.02
Tax Rate on Formal Goods Consumption, tc,	0.115
Tax Rate on Capital Income, tk	0.091

Economy, Reserve Bank of India, which is available at https://dbie.rbi.org.in. The data for the real government expenditure on schooling and school enrolment rate in classes I–VIII are obtained from various issues of Educational Statistics at a Glance, Government of India, which are available at https://www.education.gov.in/educational-statistics-glance-1. Data for school enrolment rate in tertiary education are obtained from the World Bank's World Development Indicators and the University Grant Commission report, 2008.

All variables are in real per capita terms (Base 2011–12), with the exception of school enrolment variables. To filter out trends, I take the log of observables and filter them using quadratic time-trend.[9]

This study endeavours to understand rich and poor households' dynamic behaviour of time spent schooling. No time-series data are available for time spent on schooling and training in India. In the absence of such data, I use observed fluctuations in school enrolment rate in classes I–VIII and in tertiary education to capture rich and poor households' dynamic behaviour of time spent schooling. The underlying assumption is that most of the cyclical fluctuations observed in lower grades are attributable to poor households' participation decisions, while the majority of the observed cyclical fluctuations in higher grades are attributable to rich households' participation decisions.

This assumption is consistent with existing empirical evidence on individuals' school attainment based on wealth distribution in India. The 2015 UNESCO-WIDE data show that the average number of years of schooling and school completion rate at all levels are much higher for individuals that belong to richer households than those that belong to poorer households in India (Table 1). The average number of years of schooling for individuals in the top fifth quintile of wealth was approximately 13 years, while it was just 5 years for individuals in the first quintile. Similarly, primary and lower secondary completion rates for children that belong to the top fourth and fifth quintiles were close to 100%. In contrast, primary and lower secondary completion rates for children that belong to the first quintile was just 0.80 and 0.59, respectively. Azam and Blom (2008) reveal a similar pattern using five rounds of National Survey Data on Employment and Unemployment for 1982–2004 period.

Previous research also shows considerable income inequality in access to tertiary education in India (Azam and Blom, 2008; Tilak and Chaudhary, 2021). For example, Azam and Blom (2008) find that the gross enrolment ratio in tertiary education was just 2.2% and 2% in 1993 and 2004, respectively for the first consumption expenditure quintile. The corresponding figures for the fifth consumption expenditure quintile were 18.6% and 29.1%, respectively. Tilak and Chaudhary (2021) use data from the 71st round of survey on Education in India, 2014 and reveal a similar trend. In 2013–2014, only 8.15% of individuals in the 18–23-year-old age group that belonged to the first consumption expenditure quintile attended higher (tertiary) education. The corresponding figure for individuals in the same age group that belonged to the fifth consumption expenditure quintile was 51.65%.

I log-linearise the models for estimation, obtaining model likelihoods using the Kalman filter. I use the Markov Chain-Monte Carlo method for posterior simulations, employing three chains and 600000 draws from each chain. I discard the first 50% of draws to minimise the impact of initial values. I also control for the variance in candidate distribution from which simulations are drawn to achieve an acceptance rate of around 25%. One period represents one year. Annual data are used because data on school enrolment rates and government expenditure on education are only available at an annual frequency.

3.1. Priors and calibrated parameters

As is standard, I estimate a set of parameters and calibrate the rest, and the choice of estimated parameters is guided by feasibility. I estimate the fraction of rich households (ω) elasticity of substitution between formal and informal goods (τ), labour adjustment cost parameter in the formal sector (ψ), elasticities of human capital to government investment (θ_{gR}) and (θ_{gP}) and autoregressive parameters and standard deviations of all exogenous shock processes. The remaining parameters are calibrated. To the extent possible, the values of calibrated parameters are chosen based on existing literature and to match selected long-term features of the data.

Calibration

This study features three types of calibrated parameters. The first set of parameters has values that are common across all models. The values of this set of parameters are presented in Table 2. As limited information is available on the value of the elasticity of human capital to time spent schooling (θ), I estimate the model for a wide range of θ values and compare the models' performance using Bayes' ratio, reporting results for θ = 0.05, 0.10, 0.15, 0.20, 0.25, 0.30 & 0.50. As demonstrated below, the data sample favour models with low θ values ($\leq$ 0.20).

The values of the remaining parameters vary across models and are chosen to match selected long-term data features. Tables 3 and 4 show the respective values of these parameters and calibration targets. I set β_R = 0.96; a value that is commonly used in the literature. I set β_P = 0.95 such that poor households would like to borrow in equilibrium in the neighbourhood of steady-state. The elasticity of output for private capital in the formal sector (α_f) and the depreciation rate of physical capital (δ_k) are set to 0.33 and 0.10, respectively, which are values used commonly in the literature. The depreciation rate of human capital (δ_h) is set equal to 0.01, in line with DeJong and Ingram (2001) and Dissou et al. (2016). I set ζ to 2; a value that is commonly used for the inverse Frisch-elasticity of labour supply in the literature.

The weights of leisure in the utility function (μ^R and μ^R) are set such that households spend approximately one-third of their time

[9] The correspondence between observables and model variables are as follows: real GDP = gdp_t, real private consumption expenditure = pc_t, real private physical investment expenditure = $p_{ft}I_t$, real government expenditure on schooling = $p_{ft}g_{ht}$. As discussed below, school enrolment rate in classes I–VIII = s_{Pt} and school enrolment rate in tertiary education = s_{Rt}.

Table 3
Calibrated values of parameters that differ across models.

Values of θ							
Parameter	0.05	0.10	0.15	0.20	0.25	0.30	0.50
Ah	1.60	1.15	1.10	1.10	1.20	1.20	1.20
κ_n	0.5161	0.5167	0.5174	0.5170	0.5167	0.5167	0.5165
κ_h	0.5090	0.5181	0.5275	0.5370	0.5466	0.5565	0.5976
μ_R	1.3657	1.3696	1.3297	1.2838	1.2378	1.2009	1.0661
μ_P	27.0740	26.2502	25.6149	24.9160	24.3012	23.6543	21.4073
ϕ_{P_s}	0.0242	0.0942	0.1576	0.2224	0.2881	0.3534	0.6127
ϕ_{R_s}	0.0036	0.0518	0.1890	0.3363	0.4892	0.6399	1.2280
gp	0.12	0.07	0.05	0.04	0.05	0.023	0.003
gc	0.60	0.10	0.26	0.195	0.16	0.114	0.0147
gh	0.32	0.18	0.14	0.105	0.086	0.062	0.008
vr	1.45	1.40	1.10	0.84	0.68	0.48	0.063
vp	0.36	0.35	0.275	0.21	0.17	0.12	0.016

Table 4
Calibration targets.

Variables	Target	Models
Share of Informal Sector in GDP	0.50	0.50
Share of Informal Sector in Household Employment	0.86	0.86
Formal and Informal Wage Ratio	3.50	3.50
Govt. Education Expenditure to GDP Ratio	0.038	0.038
Govt. Consumption Expenditure to GDP Ratio	0.07	0.07
Pvt. Education Expenditure to GDP Ratio	0.015	0.015
Ratio of Poor to Rich Households' Education Expenditure of	0.17	0.17
Fraction of Time Spent Working by Households	0.33	0.33
Fraction of Time Spent Schooling by Poor Households	0.07	0.07
Fraction of Time Spent Schooling by Rich Households	0.10	0.10
Ratio of Poor to Rich Households' Human Capital	0.50	0.50
Ratio of Consumption Tax Revenue to GDP	0.061	0.061
Ratio of Income Tax Revenue to GDP	0.007	0.007
Ratio of Corporate Tax Revenue to GDP	0.015	0.015
Ratio of Poor Households' Transfer Payment to GDP	0.01	0.01

working at the previous/calibrated values of other parameters. No systematic evidence is available on the amount of time spent schooling and training by individuals in India. For the US, DeJong and Ingram (2001) find that workers spend one-third of their work time on schooling and training. In the absence of data from India, I set ϕ_{R_s} such that rich households spend approximately one-third of their work time on schooling ($s_R = 0.10$) in the steady-state.

As previously noted, educational attainment of individuals belonging to the top two wealth quintiles is substantially higher than of individuals in lower wealth quintiles in India. To reflect this difference for the purpose of calibration and setting priors, I assume that the proportion of rich households is 40% ($\omega = 0.40$) (prior mean).[10] Data from India show that individuals' average number of years of education in the second and fourth wealth quintiles in 2015 were 7.50 years and 10.75 years, respectively. I set ϕ_{P_s} such that the time spent by poor households is ($\frac{7.50}{10.75} \approx 0.70$) of the time spent by rich households on schooling and training ($s_P = 0.07$) in the steady-state.

To reflect the lower quality of education and training received by poor households, I set κ_h such that poor households' steady-state human capital stock is half of the rich households. Using national accounts data, Tilak (2009) and Motkuri and Revathi (2023) estimate that the private schooling expenditure to GDP ratio averaged 1.5% in India over 1990–2017 period. Consistent with their estimates, I set the unit cost of schooling for rich (v_R) and poor (v_P) households such that the private schooling expenditure to GDP ratio is 1.5% in the steady-state. In addition, to reflect poor households' lower schooling cost, I set v_P to one-fourth of v_R. This implies that poor households' total schooling expenditure is approximately one-sixth that of rich households, which is consistent with the estimates of Sengupta et al. (2008) and Tilak (2009) based on the 55th and 61st rounds of NSSO Survey on Employment and Unemployment for the period 1999–2000 and 2004–2005.

Empirical evidence demonstrates that the informal sector accounts for 86% of total employment and approximately 50% of GDP in India (NCEUS, 2007; GOI, 2012). In addition, the average wage in the formal sector is much higher than in the informal sector. Using data from two rounds of India's National Sample Survey for 2004–2005 and 2011–2012, Kumar and Pandey (2021) find that the average wage in the formal sector was approximately 3.5 times larger than the average wage in the informal sector. Therefore, I set the informal sector' s relative productivity (κ_n), the elasticity of output with respect to rich households' effective labour in the informal sector (α_n), the share of formal goods in consumption (χ) and the relative disutility for rich households from working in the formal sector (ϕ_{lf}) such that the models match the share of informal employment in total employment (0.86) and GDP (0.50) and the formal and informal sector wage earnings differential (3.50) at the previous/calibrated values of other parameters.[11]

The data reveals that government expenditure on food subsidies averaged 1% in India from 1990 to 2016. I set the value of gp such that poor households' total transfer equals 1% of GDP. The values of remaining calibrated parameters are set to match the steady-state values of macro-aggregates in India. I set the steady-state levels of government schooling investment (gh) and government consumption expenditure (gc) to match average government school expenditure

[10] This is also consistent with evidence on financial inclusion in India. Using the World Banks' Global Findex Database, which is available at https://www.worldbank.org/en/publication/globalfindex/Data, I find that the average proportion of individuals 15 years of age and above with a bank account in India was 41% over the 1992–2019 sample period.

[11] $\phi_{lf} > 0$ is required because rich households are indifferent concerning working in formal and informal sectors at the margin. The tax on wage income in India is not large enough to generate an observed wage differential between these two sectors. Therefore, ϕ_{lf} can capture the disutility incurred from obtaining formal jobs in a reduced form.

to GDP ratio (0.038) and government final consumption expenditure net of schooling expenditure to GDP ratio (0.07) for the 1992–2019 period.[12] I set wage tax (tw), sales tax (tc) and capital tax (tk) rates to match the average income tax revenue to GDP ratio (0.007), sales tax revenue to GDP ratio (0.061) and corporate tax revenue to GDP ratio (0.015), respectively. Finally, as distortionary tax rates are fixed and only rich households' lump-sum taxes/transfers adjust in response to shocks (poor households' income transfers are governed by an exogenous process that is independent of other shocks), constrained Ricardian equivalence holds in the model in the sense that government expenditure financing by either lump-sum tax or government borrowing does not affect equilibrium outcomes (see the equilibrium equations in the Appendix). Therefore, I assume that any shortfall in government revenue is financed via lump-sum tax (to_t), setting government borrowing (b_{t+1}) to zero for all t.

Priors

Table 5 presents the priors for the parameters. In choosing priors, I largely follow previous literature. I choose Beta distribution with support $[0,\ 1]$ for ω and autoregressive coefficients (ρ_i) and set the prior standard deviation to 25% of their prior mean. I assume that the priors for standard deviations of structural shocks (σ_i) follow an extremely diffused inverse-gamma distribution, setting their standard deviation to ∞. I also assume that θ_{gR}, θ_{gP} and τ follow a diffused normal distribution with a prior standard deviation equal to half of the prior mean. ψ is assumed to follow a gamma distribution with a prior standard deviation equal to half of the prior mean.

As noted above, I set the prior mean of $\tau = 8$, as in Fernández and Meza (2015) and Kumar (2023). The prior mean of ω is 0.40. No evidence is available regarding elasticities of human capital to public expenditure (θ_{gR} and θ_{gP}) for India. Similar to the elasticity of human capital to time spent schooling (θ), I experiment with various values of θ_{gR} and θ_{gP}, setting the prior mean equal to θ, with a standard deviation that is half of the prior mean. For example, in the model with $\theta = 0.05$, I set a prior mean of θ_{gR} & $\theta_{gP} = 0.05$, with the standard deviation of 0.025. Similarly, in the model with $\theta = 0.10$, I set a prior mean of θ_{gR} & $\theta_{gP} = 0.10$, with a standard deviation of 0.05.

I set $\rho_A = 0.41$ and $\sigma_A = 0.013$, referring to the values estimated by Ghate et al. (2016) for the TFP process in India for the 1992–2010 period. To set the prior mean of autoregressive coefficients ρ_{gh} and ρ_{gc} and their standard deviations (σ_{gh} and σ_{gc}), I estimate the first-order autoregressive process for these variables using filtered data for the 1992–2019 period, setting their priors equal to estimated values. Similarly, to set the prior mean and standard deviation of ρ_{gp} and σ_{gp}, I estimate the first-order autoregressive process for filtered data of the real food subsidy expenditure in India for 1990–2016 period. Accordingly, I set the prior mean of ρ_{gp} and σ_{gp} to 0.59 and 0.018, respectively.

Finally, I set ψ, ρ_{Ahp}, ρ_{Ahr}, ρ_{vp}, ρ_{vr}, ρ_{γ}, σ_{Ahp}, σ_{Ahr}, σ_{vp}, σ_{vr} and σ_{γ} such that the models match the persistence and volatility of private consumption, private investment, school enrolment rate in classes I–VIII and school enrolment rate in tertiary education at calibrated/prior mean values of parameters.

4. Posterior estimation

Tables 6A–6D present the estimation results. I report the posterior mean, the 90% credible interval of models' estimated parameters and their log marginal densities (Modified Harmonic mean).

The results reveal that the model with $\theta = 0.15$ has the highest log marginal density (328.88), followed by the model with $\theta = 0.10$ (328.03), and $\theta = 0.20$ (327.50). Based on a comparison of log marginal data densities (Kass and Raftery, 1995), I find that the data favour the model with $\theta = 0.15$ over models with $\theta = 0.05$ (326.68) and $\theta = 0.25$ (326.20).[13] The data strongly favour the model with $\theta = 0.15$ over models with $\theta = 0.30$ (301.94) and $\theta = 0.50$ (309.74). In summary, the results demonstrate that the elasticity of human capital to time spent schooling is relatively low in India.

Considering these results, I focus on models with $\theta = 0.10,\ 0.15$ & 0.20 for the remainder of the study. The results reveal that the posterior mean and 90% credible interval of the parameters are similar across all three models except for the elasticities of human capital to public expenditure (θ_{gR} and θ_{gP}). The posterior mean of θ_{gP} in the model with $\theta = 0.15$ is 0.1472 compared with 0.1086 and 0.1677 for the models with $\theta = 0.10$ and $\theta = 0.20$, respectively. However, their 90% credible intervals overlap, indicating that these differences are insignificant.

The estimates of posterior mean of θ_{gR} are similar across models with a posterior mean of $\approx 0.08-0.10$. For all three models, the posterior mean of θ_{gP} is larger than the posterior mean of θ_{gR}, indicating that public schooling investment has a larger effect on poor households' human capital than that of rich households. The 90% credible interval of θ_{gP} is strictly positive; however, the 90% credible interval of θ_{gR} contains zero for models with $\theta = 0.15$ & 0.20, indicating an insignificant effect of public schooling investment for rich households.

These results are consistent with the evidence in India that the public schooling investment has focused on expanding primary and secondary schooling, particularly since the 1990s (Duraisamy and Duraisamy, 2016; Prakash, 2007). The expansion in higher education is primarily attributable to the private sector. For example, Duraisamy and Duraisamy (2016) find that enrolment in unaided private higher education institutions increased from 25% in 2000–2001 to 58% in 2012–2013.

The posterior mean of $\omega \approx 0.73 - 0.75$ is much larger than its prior mean (= 0.40), which is consistent with the findings of Gabriel et al. (2016), who estimate a NK model with non-Ricardian households for India. The estimated posterior mean for the proportion of Ricardian households in their model is 0.80. Kumar (2023) also finds that the NK model with a low share of non-Ricardian households outperforms models with a high share of non-Ricardian households.

Similarly, the posterior mean of $\tau \approx 12$ is higher than its prior mean (= 8), indicating a higher degree of substitutability between formal and informal goods in consumption. The results are not very informative concerning the adjustment cost parameter (ψ) as it is not precisely estimated.

Estimation results show that all shocks are highly persistent ($\rho_i > 0.50$), with the exception of rich households' TFP and tuition cost. The posterior 90% intervals of parameters of shocks processes are much narrower than their prior 90% interval, indicating that the data for these parameters are extremely informative.

Matching moments

Table 7 compares the cross-correlations, auto-correlations and volatility of the key macroeconomic and schooling variables generated by estimated models relative to the data.[14] The results reveal that the models well match the volatility of GDP, private consumption and physical capital investment, public schooling investment and poor households' time spent schooling; however, they overestimate the

[12] The government' s final consumption expenditure reported in the national accounts includes current expenditure on schools such as wages and salaries for teachers and staff. In India, around 94% of government schooling expenditure is current expenditure, with the largest proportion being wages and salaries for teachers and staff. To be consistent with the model, I separate government schooling expenditure from government final consumption expenditure.

[13] Taking model X as null and model Y as alternative, based on 2 ln(Bayes Factor) statistics Kass and Raftery (1995) suggest following criteria. If 2 ln(Bayes Factor) > 10 yields extremely strong evidence against the null, a value between 6 and 10 is strong evidence against the null, a value between 2 and 6 is positive evidence against the null and if it is between 0 and 2, it is insignificant.

[14] The values of estimated parameters are set at their posterior mean.

Table 5
Prior distributions for parameters.

Parameter	Distribution	Range	Mean	S.D.	90% Interval
ω	Beta	$R+$	0.40	0.10	0.2722, 0.5318
τ	Normal	$[-\infty, +\infty]$	8.00	4	2.8738, 13.1262
ψ	Gamma	$R+$	0.50	0.25	0.2181,0.8351
Elasticity of Human Capital to Govt. Inv. in Schools, $\theta_{gP}\&\theta_{gR}$					
Parameter	Distribution	Range	Mean	S.D.	90% Interval
$\theta_{gP}\&\theta_{gR}$	Normal	$[-\infty, +\infty]$	0.05	0.025	0.0180, 0.0820
$\theta_{gP}\&\theta_{gR}$	Normal	$[-\infty, +\infty]$	0.10	0.05	0.0359, 0.1641
$\theta_{gP}\&\theta_{gR}$	Normal	$[-\infty, +\infty]$	0.15	0.075	0.0539, 0.2461
$\theta_{gP}\&\theta_{gR}$	Normal	$[-\infty, +\infty]$	0.20	0.10	0.0718, 0.3282
$\theta_{gP}\&\theta_{gR}$	Normal	$[-\infty, +\infty]$	0.25	0.125	0.0718, 0.3282
$\theta_{gP}\&\theta_{gR}$	Normal	$[-\infty, +\infty]$	0.30	0.15	0.1078, 0.4922
$\theta_{gP}\&\theta_{gR}$	Normal	$[-\infty, +\infty]$	0.50	0.25	0.1796, 0.8204
Autoregressive Coefficients					
Parameter	Distribution	Range	Mean	S.D.	90% Interval
ρ_A	Beta	[0,1]	0.41	0.1025	0.2722, 0.5318
ρ_{Ahp}	Beta	[0,1]	0.62	0.155	0.4087, 0.8177
ρ_{Ahr}	Beta	[0,1]	0.62	0.155	0.4087, 0.8177
ρ_{gh}	Beta	[0,1]	0.60	0.15	0.3972, 0.7926
ρ_{gc}	Beta	[0,1]	0.62	0.15	0.4087, 0.8177
ρ_{gp}	Beta	[0,1]	0.59	0.15	0.3914, 0.7799
ρ_{vr}	Beta	[0,1]	0.50	0.1205	0.3364, 0.6636
ρ_{vp}	Beta	[0,1]	0.70	0.130	0.5222, 0.8609
ρ_γ	Beta	[0,1]	0.50	0.1205	0.3364 0.6636
Standard Deviation of Shocks					
Parameter	Distribution	Range	Mean	S.D.	90% Interval
σ_A	Inverse-Gamma	R^+	0.013	∞	0.0033, 0.0244
σ_{Ahp}	Inverse-Gamma	R^+	0.023	∞	0.0059, 0.0432
σ_{Ahr}	Inverse-Gamma	R^+	0.023	∞	0.0059, 0.0432
σ_{gh}	Inverse-Gamma	R^+	0.055	∞	0.0141, 0.1034
σ_{gc}	Inverse-Gamma	R^+	0.051	∞	0.0131, 0.0959
σ_{gp}	Inverse-Gamma	R^+	0.018	∞	0.0046, 0.0338
σ_{vr}	Inverse-Gamma	R^+	0.035	∞	0.0090, 0.0658
σ_{vp}	Inverse-Gamma	R^+	0.04	∞	0.0103, 0.0752
σ_γ	Inverse-Gamma	R^+	0.01	∞	0.0026, 0.0188

Table 6A
Posterior estimation.

Parameter	$\theta = 0.05$		$\theta = 0.10$	
	Mean	90% Interval	Mean	90% Interval
ω	0.7481	0.6619, 0.8365	0.7453	0.6574, 0.8366
τ	12.3511	7.3077, 17.0179	12.3683	7.4066, 17.1191
ψ	0.5094	0.1239, 0.8773	0.5083	0.1212, 0.8733
θ_{gP}	0.0542	0.0142, 0.0953	0.1086	0.0333, 0.1814
θ_{gR}	0.0466	0.0059, 0.0876	0.0803	0.0003, 0.1603
ρ_A	0.4164	0.3180, 0.5166	0.4147	0.3173, 0.5140
ρ_{Ahp}	0.6157	0.4069, 0.8266	0.6169	0.4081, 0.8169
ρ_{Ahr}	0.5637	0.3317, 0.8013	0.5628	0.3283, 0.8018
ρ_{gh}	0.5526	0.3976, 0.7125	0.5558	0.3990, 0.7167
ρ_{gc}	0.7476	0.6685, 0.8282	0.7458	0.6662, 0.8283
ρ_{gp}	0.5908	0.3572, 0.8407	0.5907	0.3538, 0.8365
ρ_{vr}	0.4908	0.3303, 0.6510	0.4885	0.3278, 0.6515
ρ_{vp}	0.5252	0.3374, 0.7097	0.5253	0.3387, 0.7136
ρ_γ	0.5488	0.4301, 0.6688	0.5483	0.4307, 0.6715
σ_A	0.0101	0.0091, 0.0112	0.0102	0.0091, 0.0112
σ_{Ahp}	0.0188	0.0077, 0.0286	0.0184	0.0079, 0.0274
σ_{Ahr}	0.0153	0.0054, 0.0259	0.01555	0.0055, 0.0260
σ_{gh}	0.0506	0.0415, 0.0591	0.0507	0.0418, 0.0595
σ_{gc}	0.1083	0.0919, 0.1246	0.1108	0.0923, 0.1284
σ_{gp}	0.0165	0.0041, 0.0305	0.0161	0.0042, 0.0296
σ_{vr}	0.0892	0.0745, 0.1044	0.0893	0.0727, 0.1051
σ_{vp}	0.0324	0.0142, 0.0476	0.0313	0.0136, 0.0468
σ_γ	0.0100	0.0086, 0.0113	0.0100	0.0086, 0.0114
Log-Density	326.68		328.03	

Table 6B
Posterior estimation.

Parameter	$\theta = 0.15$		$\theta = 0.20$	
	Mean	90% Interval	Mean	90% Interval
ω	0.7387	0.6460, 0.8337	0.7358	0.6430, 0.8322
τ	12.2558	7.4789, 17.0003	12.4744	7.5824, 17.1917
ψ	0.5247	0.1276, 0.9076	0.5217	0.12304, 0.8970
θ_{gP}	0.1472	0.0536, 0.2759	0.1677	0.0674, 0.2679
θ_{gR}	0.0862	−0.0286, 0.2022	0.1002	−0.0397, 0.2429
ρ_A	0.4081	0.3101, 0.5091	0.4108	0.3101, 0.5096
ρ_{Ahp}	0.6079	0.4033, 0.8121	0.6087	0.4194, 0.8011
ρ_{Ahr}	0.5563	0.3228, 0.7978	0.5571	0.3272, 0.7913
ρ_{gh}	0.5606	0.4077, 0.7214	0.5591	0.4021, 0.7201
ρ_{gc}	0.7468	0.6670, 0.8288	0.7430	0.6629, 0.8257
ρ_{gp}	0.5862	0.3473, 0.8345	0.5895	0.3482, 0.8304
ρ_{vr}	0.4889	0.3276, 0.6498	0.4854	0.3216, 0.6474
ρ_{vp}	0.5235	0.3410, 0.7077	0.5217	0.3356, 0.7116
ρ_γ	0.5506	0.4356, 0.6718	0.5483	0.4315, 0.6704
σ_A	0.0102	0.0091, 0.0112	0.0102	0.0091, 0.0112
σ_{Ahp}	0.0169	0.0076, 0.0252	0.0177	0.0085, 0.0256
σ_{Ahr}	0.0156	0.0053, 0.0261	0.0156	0.0055, 0.0263
σ_{gh}	0.0509	0.0420, 0.0598	0.0508	0.0420, 0.0597
σ_{gc}	0.1144	0.0927, 0.1369	0.1204	0.0949, 0.1450
σ_{gp}	0.0194	0.0039, 0.0373	0.0173	0.0040, 0.0319
σ_{vr}	0.0792	0.0655, 0.0942	0.1021	0.0830, 0.1227
σ_{vp}	0.0287	0.0138, 0.0417	0.0331	0.0131, 0.0520
σ_γ	0.0101	0.0086, 0.0115	0.0101	0.0087, 0.0115
Log-Density	328.88		327.50	

Table 6C
Posterior estimation.

Parameter	$\theta = 0.25$		$\theta = 0.30$	
	Mean	90% Interval	Mean	90% Interval
ω	0.7242	0.6234, 0.8240	0.2668	0.0763, 0.4963
τ	12.4943	7.6609, 17.2779	1.7641	−0.0784, 3.8830
ψ	0.5127	0.1252, 0.8777	0.4250	0.1042, 0.7063
θ_{gP}	0.1830	0.0840, 0.2828	0.0791	−0.0405, 0.1992
θ_{gR}	0.0808	−0.0748, 0.2453	0.5903	0.4596, 0.7293
ρ_A	0.4091	0.3086, 0.5096	0.4885	0.3513, 0.6046
ρ_{Ahp}	0.6044	0.4230, 0.7895	0.6399	0.5001, 0.8042
ρ_{Ahr}	0.5540	0.3236, 0.7900	0.5720	0.3682, 0.8536
ρ_{gh}	0.5623	0.4073, 0.7219	0.5459	0.4065, 0.6801
ρ_{gc}	0.7412	0.6609, 0.8241	0.8073	0.7413, 0.8578
ρ_{gp}	0.5623	0.4073, 0.7219	0.6003	0.4615, 0.7605
ρ_{vr}	0.4802	0.3180, 0.6426	0.5260	0.3747, 0.6869
ρ_{vp}	0.5152	0.3250, 0.7061	0.4864	0.2824, 0.6705
ρ_γ	0.5457	0.4274, 0.6663	0.5004	0.3143, 0.7020
σ_A	0.0102	0.0092, 0.0113	0.0078	0.0067, 0.0089
σ_{Ahp}	0.0177	0.0092, 0.0249	0.0180	0.0079, 0.0263
σ_{Ahr}	0.0159	0.0054, 0.0263	0.0279	0.0058, 0.0606
σ_{gh}	0.0509	0.0419, 0.0598	0.0451	0.0366, 0.0541
σ_{gc}	0.1369	0.1015, 0.1725	0.0615	0.0374, 0.0904
σ_{gp}	0.0160	0.0041, 0.0302	0.2679	0.0647, 0.5174
σ_{vr}	0.1156	0.0901, 0.1444	0.1471	0.0412, 0.2389
σ_{vp}	0.0340	0.0122, 0.0549	0.0441	0.0105, 0.0964
σ_γ	0.0100	0.0086, 0.0114	0.0108	0.0034, 0.0207
Log-Density	326.20		301.94	

volatility of rich households' time spent schooling.

The models also match the persistence of poor and rich households' GDP, private physical capital investment, public schooling investment and time spent schooling, overestimating the persistence of private consumption. Similar to the data, the models generate pro-cyclical private physical investment, poor households' time spent on schooling and rich households' acyclical time spent schooling. They overestimate the pro-cyclicality of private consumption. The models also generate acyclical public schooling investment, while it is marginally pro-cyclical in the data.

Variance decomposition

Table 8 presents the variance decomposition based on the model with $\theta = 0.15$. For brevity, I do not report the variance decomposition based on models with $\theta = 0.10$ and $\theta = 0.20$; however, the results are very similar.

The results reveal that the majority of variation in GDP, formal and informal sectors' output, private physical capital investment and wages in formal and informal sectors can be explained by TFP shocks. TFP shocks also account for the plurality of variation in private consumption expenditure. The other significant drivers of private consumption expenditure include government consumption expenditure (30%) and preference (20%) shocks.

Variations in poor households' time spent schooling are mainly attributable to shocks to poor schools' productivity (72%), public schooling investment (10%), poor households' tuition cost (7%) and preference shocks (10%). In contrast, variations in rich households' time spent schooling are overwhelmingly attributable to tuition cost shocks (96%). Shocks to poor schools' productivity account for the majority

Table 6D
Posterior estimation.

Parameter	$\theta = 0.50$	
	Mean	90% Interval
ω	0.6103	0.4651, 0.7553
τ	11.3681	6.3479, 16.1573
ψ	0.8026	0.1900, 1.4348
θ_{gP}	0.1861	0.1059, 0.2610
θ_{gR}	−0.0283	−0.1955, 0.0997
ρ_A	0.3867	0.2863, 0.4888
ρ_{Ahp}	0.5669	0.4225, 0.7092
ρ_{Ahr}	0.3988	0.1885, 0.6608
ρ_{gh}	0.5794	0.4206, 0.7368
ρ_{gc}	0.7049	0.6172, 0.7959
ρ_{gp}	0.5884	0.3385, 0.8269
ρ_{ur}	0.4712	0.2747, 0.6754
ρ_{up}	0.5035	0.2926, 0.7091
ρ_{γ}	0.5277	0.4123, 0.6411
σ_A	0.0103	0.0092, 0.0114
σ_{Ahp}	0.0188	0.0160, 0.0213
σ_{Ahr}	0.0408	0.0065, 0.0660
σ_{gh}	0.0511	0.0419, 0.0600
σ_{gc}	1.1509	0.5539, 1.8466
σ_{gp}	0.0156	0.0040, 0.0280
σ_{ur}	1.3324	0.0067, 4.1959
σ_{up}	0.0404	0.0091, 0.0767
σ_{γ}	0.0096	0.0084, 0.0109
Log-Density	309.74	

Table 7
Matching moments.

Standard Deviation				
Variable	Data	$\theta = 0.10$	$\theta = 0.15$	$\theta = 0.20$
gdp_t	0.021	0.0144	0.0144	0.0149
pc_t	0.012	0.0133	0.0135	0.0136
$p_{ft}I_t$	0.062	0.0659	0.0656	0.0650
$p_{ft}gh_t$	0.066	0.0608	0.0611	0.0609
s_{Pt}	0.041	0.0384	0.0386	0.0407
s_{Rt}	0.047	0.1073	0.1074	0.1051
First-Order Auto-Correlation				
Variable	Data	$\theta = 0.10$	$\theta = 0.15$	$\theta = 0.20$
gdp_t	0.517	0.5761	0.5694	0.5973
pc_t	0.213	0.7741	0.7757	0.7795
$p_{ft}I_t$	0.538	0.4300	0.4348	0.4300
$p_{ft}gh_t$	0.604	0.5579	0.5619	0.5513
s_{Pt}	0.617	0.5596	0.5451	0.5585
s_{Rt}	0.381	0.5037	0.5039	0.4693
Correlation with gdp_t				
Variable	Data	$\theta = 0.10$	$\theta = 0.15$	$\theta = 0.20$
pc_t	0.312	0.6173	0.6218	0.6592
pf_tI_t	0.876	0.7433	0.7456	0.7348
$p_{ft}gh_t$	0.313	0.0223	−0.0123	−0.0023
s_{Pt}	0.396	0.2403	0.2369	0.1622
s_{Rt}	−0.097	0.0594	0.0759	0.0221

of variations in poor households' labour supply (63%), followed by tuition cost shocks. Variations in the formal sector's labour supply are primarily attributable to shocks to TFP, government consumption expenditure and preference.

Overall the results demonstrate that while TFP shocks account for the majority of variations in GDP, formal and informal sector GDPs and wages, variations in schooling and human capital are predominantly attributable to shocks to the costs of schooling and human capital accumulation technology in India. The result that TFP shocks account for most of the variations in GDP and output are similar to the results of DeJong and Ingram (2001) for the US.

Impulse response

Figs. 1–4 illustrate the Bayesian impulse responses to shocks to public expenditure on schooling and tuition costs based on model with $\theta = 0.15$ and 90% confidence bands. The impulse responses for models with $\theta = 0.10$ & 0.20 are very similar. Figs. 1 and 4 depict impulse responses to a one *s.d.* positive shock to government schooling investment and transfer to poor households, respectively. Figs. 2 and 3 display impulse responses to a one *s.d.* positive shock to rich and poor households' tuition costs, respectively.

Fig. 1 shows that a government schooling investment shock has an extremely persistent, non-linear effect on aggregate and sectoral GDPs, wherein aggregate and sectoral GDPs rise on impact in response to a positive shock. However, in subsequent periods, aggregate and formal sector GDPs initially fall below their steady-states and then rise above them before converging back to the steady-states. Conversely, the informal sector GDP continues to rise until the third period before falling, resulting to an inverted U-shaped response. The impulse response also indicates that the informal sector GDP response remains significant for an extended period of time, while the aggregate and formal sector GDP response becomes insignificant after the third period.

Sectoral wage and employment income responses mirror those of sectoral GDPs, with rich households' formal sector wage and employment income exhibiting a U-shaped response. Poor households' informal sector wage and employment income, which is defined as $ein_{Pt} = w_{nt}l_{Pt}h_{Pt}$, exhibit inverted U-shaped responses. Poor households' employment income remains elevated and significantly differs from zero for an extended period of time, while rich households' employment income remains significantly below its steady-state up to the tenth period. The other notable feature is that the formal wage declines significantly, but the response of informal sector wage is not significant.

A positive shock to government schooling investment raises rich and poor households' time spent schooling and human capital. Its effect on human capital is highly persistent, with human capital remaining above its steady-state for an extremely long period of time, particularly for poor households. Poor households' schooling and human capital respond much more strongly than that of rich households. The confidence bands reveal that rich households' response of time spent schooling is insignificant. This differential response is attributed to poor households' estimated mean elasticity of human capital to government human capital investment that is larger than that of rich households, $\theta_{gP} > \theta_{gR}$.

The inverted U-shaped and highly persistent response of human capital is the result of two counteracting effects. An increase in time spent schooling and government schooling investment raises human capital, but a depreciation reduces it. The first effect initially dominates, resulting in a rise of human capital in earlier periods. Subsequently, the second effect dominates; however, the low depreciation rate results in a relatively slow convergence to the steady-state.

Rich and poor households' labour supply and effective labour supply responses differ, where rich households' labour and effective labour supply increase sharply on impact. In contrast, poor households' labour supply falls. The decreased labour supply reduces poor households' effective labour supply marginally but insignificantly on impact, despite a rise in human capital.

The impulse responses also reveal that a rise in government schooling investment has a negative impact on rich households' physical capital investment. Rich and poor households' private consumption and aggregate private consumption fall on impact. However, in subsequent periods, poor households' consumption rises above the steady-state significantly before converging back to the steady-state.

An increase in government schooling investment raises the marginal productivity of time spent schooling, inducing rich and poor households to spend more time schooling and reducing their labour supply. However, increased government expenditure must be financed, which raises the tax burden. This extra expenditure is financed through raising

Table 8
Posterior Mean Variance Decomposition in % (Model $\theta = 0.15$).

Shocks	gdp_t	y_{ft}	y_{nt}	pc_t	$p_{ft}I_t$	sp_t	sr_t	l_{Pt}	l_{Rft}	w_{ft}	w_{nt}
e_{At}	89.59	85.37	70.89	41.63	58.58	0.28	0.19	0.00	25.67	72.79	98.49
e_{RAht}	2.26	2.62	0.17	2.71	0.24	0.00	2.83	0.00	2.63	2.35	0.37
e_{PAht}	0.47	0.03	25.13	0.92	0.00	71.91	0.00	63.03	0.42	0.06	0.35
e_{ght}	0.30	0.14	3.60	0.84	0.86	9.59	0.17	8.41	0.52	0.94	0.01
e_{gct}	1.75	2.60	0.57	29.73	10.51	0.00	0.36	0.00	29.22	12.81	0.10
e_{gpt}	0.00	0.00	0.01	0.00	0.00	0.00	0.00	2.63	0.00	0.00	0.00
e_{vrt}	1.41	1.81	0.12	1.73	0.11	0.00	96.35	0.00	15.47	1.86	0.24
e_{vpt}	0.00	0.00	0.07	0.00	0.00	7.27	0.00	16.51	0.00	0.00	0.00
e_{τ}	5.04	7.29	0.29	20.38	26.72	9.46	0.55	8.42	25.44	8.98	0.19

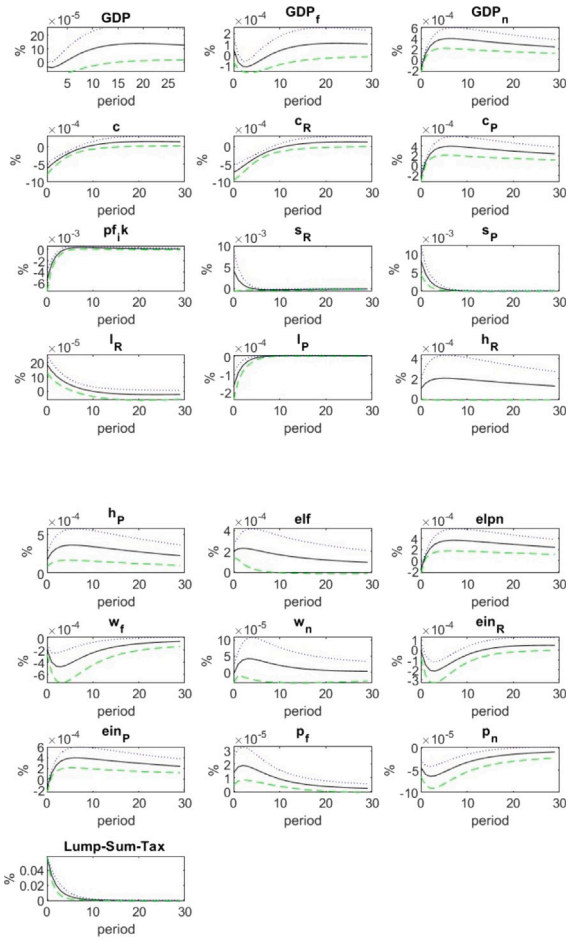

Fig. 1. Effects of a positive government schooling investment shock (in %).

the lump-sum taxes (*to*), which increases significantly, with a negative wealth effect on rich households, which induces them to raise their labour supply. The net result is that rich households' labour supply rises. The negative wealth effect and increased schooling expenditure reduces rich households' consumption. A rise in schooling expenditure also reduces poor households' consumption on impact, despite a rise in employment income.

Higher government schooling investment makes households' human capital investment more attractive. Therefore, rich households' physical capital investment falls because of the negative wealth effect and rebalancing expenditure towards human capital. Higher government schooling investment also raises the relative demand and relative price of formal goods. In contrast, greater output in the informal sector lowers the price of informal goods.

The increase in rich households' effective labour supply reduces the formal wage on impact. In subsequent periods, as rich households' effective labour supply falls it has a positive impact on formal wages. However, a fall in physical capital investment reduces the marginal productivity of formal (effective) labour, which has a negative impact on formal wages. The second effect dominates initially resulting in the U-shaped response.

The rise in rich household' labour supply increases aggregate and sectoral GDPs on impact. In subsequent periods, a rise in human capital (particularly that of poor households) results in an inverted U-shaped and highly persistent response of informal sector GDP. After the initial rise, the formal sector GDP falls below its steady-state, mainly due to a fall in the physical capital stock despite a rise in rich households' human capital. However, after 12 periods, the second effect begins to dominate and the formal sector GDP rises above its steady-state (although insignificantly) before converging back to its steady-state. The response of aggregate GDP mirrors the interaction between informal and formal sector GDP responses.

Employment income depends on wages and effective labour supply. The time path of poor households' employment income reflects the time path of their effective labour as the informal wage changes relatively little. A rise in rich households' effective labour supply and a (marginal) decline in poor households' effective labour have positive effects on informal wage. In contrast, a fall in the price of informal goods has a negative effect on the informal wage. As a result, informal wages change minimally.

Rich households' employment income time path qualitatively mirrors that of formal wage. Quantitatively, rich households' employment income falls less relative to formal wage as its decline is initially moderated by increased labour supply and a subsequent rise in human capital.

A positive shock to rich schools' tuition cost increases rich households' cost of schooling and tightens budget constraints, with a persistent negative effect on time spent schooling and human capital and a persistent positive effect on labour supply (Fig. 2). Rich households' effective labour supply increases on impact, then subsequently falls. Formal wage initially falls and then rises above its steady-state. A positive shock to rich schools' tuition cost has an insignificant effect on rich households' employment income.

While a rise in rich households' effective labour supply has a positive effect on sectoral and aggregate GDPs on impact, they fall below their steady-state levels in subsequent periods before converging. Rich and poor households' physical capital investment and consumption and aggregate consumption show similar pattern. A rise in schooling cost reduces rich households' schooling expenditure, inducing them to increase physical capital investment and consumption on impact. However, as rich households' human capital declines, the marginal rate of return on physical capital investment falls due to complementarity between physical and human capital, resulting in decreased physical capital investment and formal sector GDP in subsequent periods.

As rich households' effective labour supply increases, it raises the marginal product of informal labour in the informal sector, increasing poor households' informal wages and consumption, employment income and time spent on schooling on impact. However, as rich households' effective labour declines, it results in a fall in poor households' informal wage and consumption, employment income and time spent

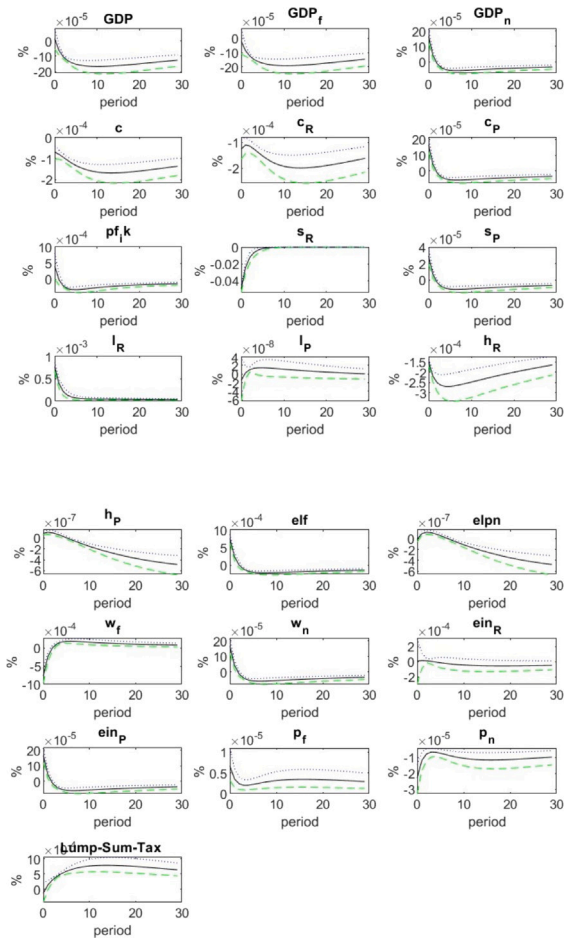

Fig. 2. Effects of a positive shock to rich households' tuition cost (in %).

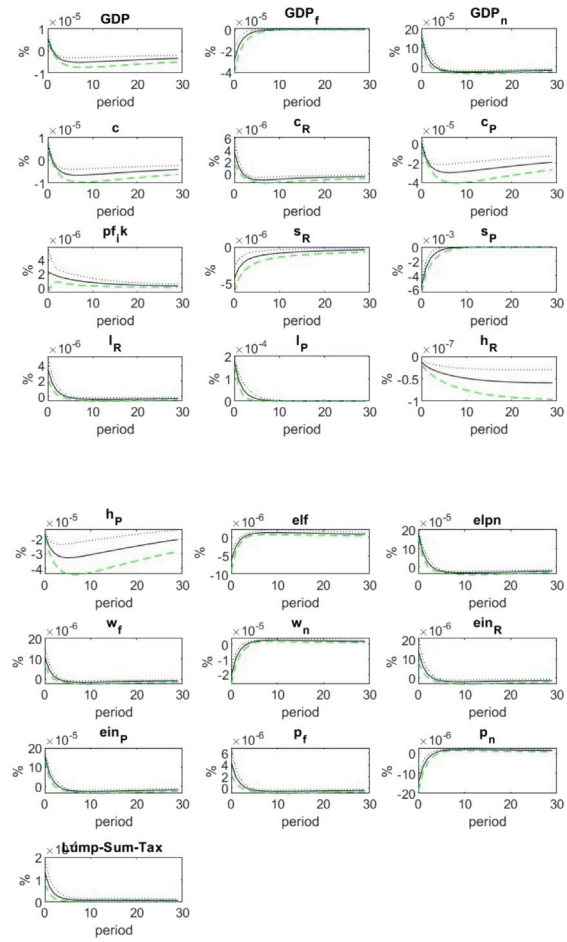

Fig. 3. Effects of a positive shock to poor households' tuition cost (in %).

schooling in subsequent periods. Therefore, the informal sector's GDP also falls in subsequent periods, which in line with the fall in formal sector GDP, reduces aggregate GDP.

A positive shock to poor schools' tuition cost increases poor households' schooling cost and tightens budget constraints, with a persistent negative effect on time spent schooling and human capital and a persistent positive effect on labour supply (Fig. 3). Poor households' effective labour supply, employment income and consumption increase on impact, but subsequently fall below their steady-state. Informal wage initially falls and then rises above its steady-state, although the rise is insignificant.

As poor households' effective labour supply increases, it raises the marginal rate of return of rich households' labour supply in the informal sector. As rich households can work in either sector, the formal wage also rises. The increased rate of return for work induces rich households to raise their labour supply and employment income and reduce time spent schooling. Rich households' consumption and private investment increase. The rise in both types of households' effective labour increases sectoral and aggregate GDPs. The output in informal sector rises relatively more, reducing the price of informal goods and raising the price of formal goods.

Finally, an increase in the lump-sum transfer to poor households from the government reduces their labour supply and employment income (Fig. 4). Increased government lump-sum transfer has a positive effect on poor households' consumption, time spent schooling and human capital. Overall, poor households' effective labour supply falls, reducing informal sector GDP and increasing informal wage and informal goods price. Since the increased transfer must be financed, the lump-sum tax (to) on rich households rises significantly. Due to the negative wealth, rich households' time spent schooling, consumption and private physical investment fall and their labour supply rises. Rich households' effective labour supply rises, resulting in decreased formal wage and formal goods price and increased formal sector GDP. Despite a rise in rich households' effective labour supply, their employment income falls. The fall in informal sector GDP is larger than the rise in formal sector GDP, resulting in a decline in aggregate GDP. Conversely, a rise in poor households' consumption dominates the fall in rich households' consumption, resulting in increased aggregate consumption in the initial period.

5. Policy analysis

Given the estimation results, I use posterior estimates of models with $\theta = 0.10,\ 0.15\ \&\ 0.20$ to examine the income and welfare effects of government schooling investment, tuition subsidies and transfer to poor households. I analyse their short- and long-term effects on poor and rich households' GDP, aggregate private consumption, private investment and employment income using expenditure multipliers. To analyse the welfare effects of these policies, I calculate the consumption equivalent relative to households' steady-state welfare.

Tuition subsidies are introduced into the model as follows. Assume that the government provides tuition subsidy to households. Let ts_{it} for $i = P\ \&\ R$ be the tuition subsidy to the ith type of household per unit of time spent schooling. Then, the net per-unit cost of households' tuition becomes $\tilde{v}_{it} = v_{it} - ts_{it}$ for $i = R\ \&\ P$. To analyse the effects of tuition subsidies, I replace v_{it} with $\tilde{v}_{it}$ in the models except in the case of the

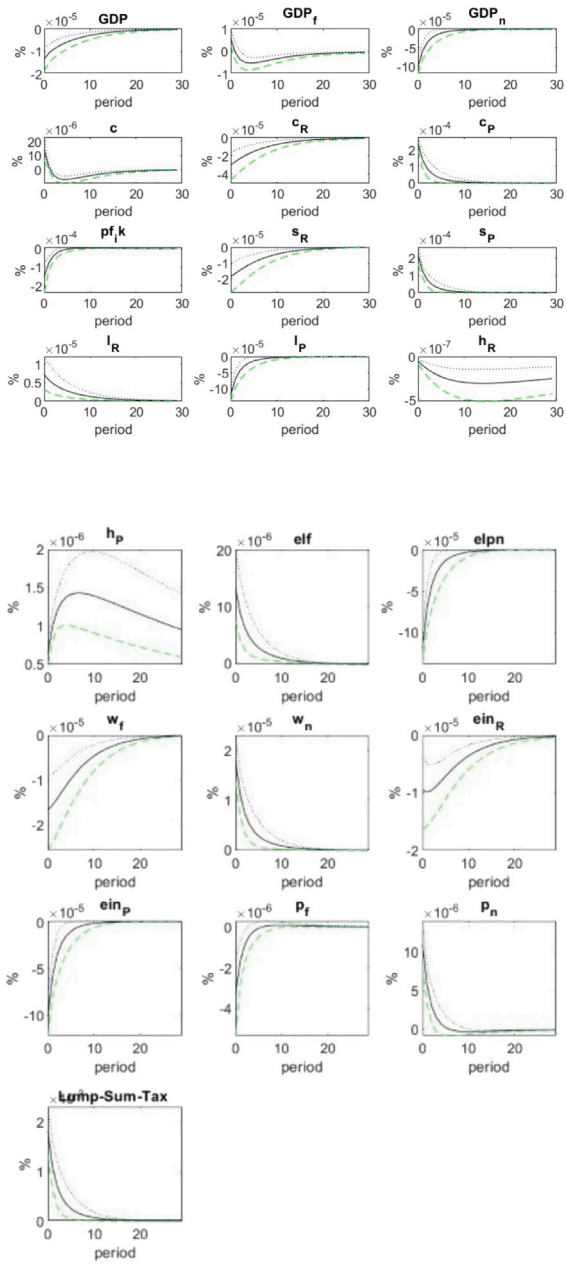

Fig. 4. Effects of a positive shock to the lump-sum transfer to the poor (in %).

market clearing condition for formal goods (2.35). The rationale for this is that tuition subsidies do not alter economy-wide resource costs. In addition, unlike changes in $v_{it}s$ that are analysed above, they must be financed, which affects rich households' decisions based on the negative wealth effect.[15]

Poor and rich households' aggregate expenditure on tuition subsidy at time t is $(1-\omega)p_{ft}s_{Pt}ts_{Pt}$ and $\omega p_{ft}s_{Rt}ts_{Rt}$, respectively. I assume that tuition subsidies in the steady-state are zero and follow the stochastic process below:

$$ts_{it} = \rho_{tsi}ts_{it-1} + e_{its} \text{ for } i = R \ \& \ P. \tag{5.1}$$

Due to the paucity of data, I do not estimate parameters of (5.1). For simulation purposes, I assume that ρ_{tsi} is uniformly distributed with the range of 0.40–0.80. I analyse two types of tuition subsidy policies, (i) Untargeted policies in which both household types receive same amount of tuition subsidy ($e_{Rts} = e_{Pts}$) and (ii) Subsidies that target either poor or rich households.

I use second-order approximation of equilibrium conditions in the neighbourhood of steady-state to calculate multipliers and consumption equivalents (Schmit-Grohé and Uribe, 2007). I then simulate selected models by independently drawing 2000 values of parameters from estimated posterior distributions. Since the model with $\theta = 0.15$ has the highest marginal density value, I use it as the benchmark model and report mean values of estimated expenditure multipliers (Tables 9–13) and consumption equivalents (Table 16) and 90% intervals. For brevity, I only report mean values of estimated expenditure multipliers (Tables 14 and 15, respectively) and consumption equivalents (Table 17) based on models with $\theta = 0.10$ & 0.20.

5.1. Government expenditure multipliers

I calculate the impact and present value government expenditure multipliers commonly used in the literature using the posterior distributions of parameters as follows:

$$\text{Impact Expenditure Multiplier} \equiv \frac{\Delta X_t}{\Delta F_t} \ \& \tag{5.2}$$

$$\text{Present Value of Expenditure Multiplier} \equiv \frac{E_t \sum_{j=0}^{k} r_{t+1}^{-j} \Delta X_{t+j}}{E_t \sum_{j=0}^{k} r_{t+1}^{-j} \Delta F_{t+j}}$$

$$\text{with } X_t = gdp_t,\ pc_t,\ p_{ft}I_t, ein_{Rt} \ \& \ ein_{Pt} \tag{5.3}$$

where F_t is the relevant government expenditure. Specifically, I calculate multipliers for total government schooling investment ($p_{ft}g_{ht}$), total transfer to poor households ($(1-\omega)p_{ft}g_{Pt}$), tuition subsidy to poor households ($(1-\omega)p_{ft}s_{Pt}ts_{Pt}$) and tuition subsidy to rich households ($\omega p_{ft}s_{Rt}ts_{Rt}$).

Multipliers of government schooling investment and transfer to poor households are calculated using changes in X_t and F_t relative to their steady-state values in response to an initial 1% rise in g_{ht} and g_{Pt} from their steady-states. For tuition subsidies, multipliers are calculated as a response to an initial 0.01-unit increase in ts_{it} from its steady-state (= 0).[16] I report present value multipliers for 10, 25 and 500 years. The 500-year period is sufficiently long to capture all the dynamics following a shock and I refer to 500-year multipliers as long-term multipliers.

Government schooling investment multipliers

Table 9 presents the multipliers of government schooling investment ($p_{ft}g_{ht}$), revealing that the government schooling investment has a significant positive long-term effect on poor households' employment income. While the mean impact of poor households' employment income multiplier is marginally negative, it rises significantly over time, with the mean 25-year and long-term multipliers of 0.77 and 1.02, respectively. The mean GDP multiplier also rises over time, with a long-term mean GDP multiplier of 0.58. However, its 90% interval does not significantly differ from zero at any horizon. The results indicate that

[15] Negative shocks to v_{it}s can be considered an improvement in the efficiency of schooling services' access and delivery such as decreased text-book and other learning material costs, improved teaching methods and lower transportation costs.

[16] For example, the impact GDP multiplier of government schooling investment is calculated as $\frac{\%\Delta gdp_t}{\%\Delta p_{ft}g_{ht}} \frac{gdp}{p_f g_h}$. The impact GDP multiplier of tuition subsidy to poor households is calculated as $\frac{\%\Delta gdp_t}{\Delta(1-\omega)p_{ft}s_{Pt}} gdp$.

Table 9
Posterior government schooling investment multipliers ($\theta = 0.15$).

Multiplier	Impact	10-Year	25-Year	Long-Term
GDP				
Mean	0.0116	0.0212	0.3257	0.5831
90% Interval	−0.0626, 0.0851	−0.1292, 0.1770	−0.1209 0.7869	−0.0790 1.2719
Pvt Con.				
Mean	−0.3478	−0.5573	−0.3323	−0.1173
90% Interval	−0.4385, −0.2696	−0.6028, −0.5100	−0.6076, 0.0675	−0.5615, 0.3583
Pvt Inv.				
Mean	−0.9029	−0.5920	−0.4643	−0.4174
90% Interval	−1.2069, −0.6150	−0.7340, −0.4631	−0.5770, −0.3517	−0.5479, −0.2777
Emp. Income				
Poor Households				
Mean	−0.0638	0.3746	0.7714	1.0197
90% Interval	−0.0920, −0.0352	0.2072, 0.5309	0.4269, 1.0994	0.5636, 1.4578
Rich Households				
Mean	−0.0012	−0.3149	−0.2906	−0.2066
90% Interval	−0.0464, 0.0408	−0.5290, −0.1371	−0.5922, 0.0138	−0.5726, 0.2034

government schooling investment has a small negative effect on rich households' employment income, which is statistically insignificant at most horizons.

The analysis reveals that increased government schooling investment has a substantial negative effect on private physical investment at all horizons, with mean impact and long-term multipliers of −0.90 and −0.42, respectively. It also has a negative effect on short-term private consumption but has an insignificant long-term impact.

As previously noted, a rise in government schooling investment results in a persistent increase in poor households' human capital, resulting in a mean long-term employment income multiplier of approximately one. However, a fall in rich households' private physical investment and consumption moderates its effect on GDP, resulting in a much smaller long-term impact.

A comparison of multipliers with models with $\theta = 0.10$ (Table 14) and $\theta = 0.20$ (Table 15) reveals that multipliers are similar across models. The only difference is that the mean long-term multipliers of poor households' GDP, private consumption and employment income are higher for the $\theta = 0.20$ model. The main reason for this is that the posterior mean of θ_{Pg} is larger and poor households' human capital responds more strongly, resulting in a larger long-term accumulation of human capital in the economy.

Tuition subsidies

Table 10 shows multipliers of tuition-subsidy targeted at poor. It shows that the mean long run employment income multiplier exceeds one (1.46). However, it has a substantially larger negative effect on GDP and employment income of poor in the short-run. The other difference is that over long term private consumption rises more in response to a tuition-subsidy to poor with the mean long run private consumption multiplier being positive (0.31). It has a negative effect on the employment income of rich with their employment income multiplier being significantly negative at all horizons.

Poor households' tuition subsidy relaxes their budget constraint and makes schooling investment more attractive. Both of these have a positive effect on poor households' time spent schooling and private consumption and a negative impact on their labour supply. The second effect dominates in the short term, resulting in a negative employment income multiplier for poor households in the initial periods. However, a long-term rise in poor households' human capital dominates, resulting in the mean long-term employment income multiplier exceeding one and a positive mean long-term private consumption multiplier.

As noted previously, since the tuition subsidy must be financed, it has a positive effect on rich households' labour supply and a negative effect on time spent schooling, private consumption and physical investment. Due to decreased formal sector wage and informal sector profit, rich households' mean employment income multiplier remains negative at all horizons, despite increased labour supply.

The tuition subsidy to rich households has a marginal effect on employment income at all horizons (Table 11). Poor households' employment income rises marginally over the long-term horizon. Its effects on GDP and private consumption are similar to that of tuition subsidies to poor households, displacing private physical capital investment relatively less over long-term horizons compared with poor households' tuition subsidy.

Rich households' tuition subsidy makes their schooling investment more attractive, inducing them to raise time spent schooling and reduce labour supply and private physical investment. The results reveal that rich households' employment income changes relatively little. Since rich households receive a tuition subsidy, no negative wealth effect is found in equilibrium. In addition, rich households' human capital rises. Both of these effects result in relatively limited decline in the private physical capital investment over longer horizons.

The untargeted tuition subsidy has a much smaller impact on poor households' employment income, with the mean long-term multiplier (0.44) that is less than half of the tuition subsidy targeting poor households (Table 12). The effects on GDP, private consumption, private capital and rich households' employment income are very similar to the tuition subsidy targeting the rich.

The effects of tuition subsidies are similar across all the three models analysed (Tables 14 and 15). Overall, the results demonstrate that among different types of tuition subsidies, targeting tuition subsidies for poor households is the most effective instrument for raising their long-term employment income.

Income transfer to poor households

An increase in poor households' income transfer has a significantly negative effect on GDP, private physical capital investment and employment income of both household types, with corresponding multipliers that are negative at all horizons (Table 13). Private consumption rises on impact, with an impact multiplier of 0.22; however, it falls over time with a long-term mean private consumption multiplier of −0.23. These patterns are similar across all models (Tables 14 and 15). One difference is that private physical capital investment declines much more on impact relative to other models, with $\theta = 0.15$. Therefore, the negative effect of income transfer to poor households on GDP, private physical capital investment and employment income is more pronounced.

5.2. Welfare costs and benefits

I calculate the welfare costs and benefits of various spending policies as a consumption equivalent relative to households' steady-state welfare. The conditional welfare (Wel^i_{jt}) at time t for rich and poor

Table 10
Posterior Tuition Subsidy Targeting Poor Households Multipliers ($\theta = 0.15$).

Multiplier	Impact	10-Year	25-Year	Long-Term
GDP				
Mean	−1.0773	−0.8493	−0.4257	−0.1092
90% Interval	−1.2595, −0.9046	−1.0048, −0.7028	−0.5406 −0.3148	−0.2136 0.0176
Pvt Con.				
Mean	−0.5227	−0.4018	−0.0033	0.3136
90% Interval	−0.6406, −0.4173	−0.4684, −0.3386	−0.0250, 0.0195	0.2480, 0.3845
Pvt Inv.				
Mean	−0.7664	−0.6247	−0.5977	−0.6034
90% Interval	−1.0029, −0.5470	−0.8089, −0.4549	−0.7740, - 0.4340	−0.7797, −0.4389
Emp. Income				
Poor Households				
Mean	−2.9635	−1.2845	0.3855	1.4622
90% Interval	−3.8938, −2.1386	−1.7068, −0.9026	0.2249, 0.5633	1.0545, 1.9385
Rich Households				
Mean	−0.3947	−0.5123	−0.4143	−0.3051
90% Interval	−0.563, −0.2457	−0.6914, −0.3461	−0.5560, −0.2789	−0.4183, −0.1915

Table 11
Posterior Tuition Subsidies Targeting Rich Households Multipliers ($\theta = 0.15$).

Multiplier	Impact	10-Year	25-Year	Long-Term
GDP				
Mean	−0.8907	−0.7981	−0.3866	−0.0483
90% Interval	−1.0563, −0.7322	−0.9510, −0.6529	−0.5003 −0.2830	−0.1708 0.0683
Pvt Con.				
Mean	−0.2877	−0.4620	−0.1853	0.0942
90% Interval	−0.3697, −0.2146	−0.5516, −0.3794	−0.2447, −0.1302	0.0179, 0.1670
Pvt Inv.				
Mean	−0.8499	−0.4758	−0.2858	−0.2024
90% Interval	−1.0739, −0.6455	−0.6073, −0.3601	−0.3913, - 0.1899	−0.3059, −0.1047
Emp. Income				
Poor Households				
Mean	−0.0717	0.0393	0.1208	0.1679
90% Interval	−0.0900, −0.0554	0.0166, 0.0653	0.0843, 0.1619	0.1197, 0.2241
Rich Households				
Mean	0.0024	−0.3057	−0.2626	−0.1564
90% Interval	−0.1278, 0.1392	−0.4145, −0.2111	−0.5398, −0.0183	−0.5570, 0.2123

Table 12
Posterior Untargeted Tuition Subsidies to Households Multipliers ($\theta = 0.15$).

Multiplier	Impact	10-Year	25-Year	Long-Term
GDP				
Mean	−0.9309	−0.8058	−0.3879	−0.0504
90% Interval	−1.0996, −0.7679	−0.9542, −0.6623	−0.4902 −0.2906	−0.1527 0.0532
Pvt Con.				
Mean	−0.3381	−0.4500	−0.1445	0.1463
90% Interval	−0.4263, −0.2578	−0.5322, −0.3720	−0.1966, −0.0946	0.0767, 0.2139
Pvt Inv.				
Mean	−0.8304	−0.5013	−0.3452	−0.2804
90% Interval	−1.0333, −0.6437	−0.6373, −0.3771	−0.4769, - 0.2320	−0.4106, −0.1634
Emp. Income				
Poor Households				
Mean	−0.6339	−0.2262	0.1799	0.4390
90% Interval	−0.7189, −0.5518	−0.3569, −0.1246	0.1263, 0.2384	0.3074, 0.5772
Rich Households				
Mean	−0.0785	−0.3457	−0.2905	−0.1825
90% Interval	−0.1643, 0.0176	−0.4749, −0.2369	−0.5277, −0.0677	−0.5076, 0.1242

households under policy regime i is obtained as follow:

$$Wel^i_{jt} = \ln c_{jt} - \mu_j \frac{(l_{jt} + \phi_{js} s_{jt})^{1+\zeta}}{1+\zeta} + \beta_j Wel_{jt+1} \text{ for } j = R \ \& \ P \tag{5.4}$$

where policy regime i refers to changes in government schooling investment, tuition subsidies or transfer to poor households as in the expenditure multiplier analysis above. Social welfare at time t is defined as follows:

$$Wel^i_t = \omega Wel^i_{Rt} + (1-\omega) Wel^i_{Pt}. \tag{5.5}$$

Corresponding steady-state welfare is obtained as follows:

$$Wel_j = \frac{\ln c_j - \mu_j \frac{(l_j + \phi_{js} s_j)^{1+\zeta}}{1+\zeta}}{1-\beta_j} \text{ for } j = R \ \& \ P. \tag{5.6}$$

Let Ω^i_j be the proportion of steady-state consumption that must be sacrificed by the jth type of household to yield the same welfare as policy regime i as follows:

$$Wel^i_{jt} = \frac{\ln(1-\Omega^i_j)c_j - \mu_j \frac{(l_j + \phi_{js} s_j)^{1+\zeta}}{1+\zeta}}{1-\beta_j} \equiv \frac{\ln(1-\Omega^i_j)}{1-\beta_j} + Wel_j. \tag{5.7}$$

Table 13
Posterior Transfer to Poor Households Multipliers ($\theta = 0.15$).

Multiplier	Impact	10-Year	25-Year	Long-Term
GDP				
Mean	−0.2686	−0.5146	−0.5818	−0.5815
90% Interval	−0.3757, −0.171	−0.7428 −0.3079	−0.8334 −0.3527	−0.8334 −0.3517
Pvt Con.				
Mean	0.2171	−0.1401	−0.2319	−0.2317
90% Interval	0.1011, 0.3439	−0.1957, −0.0914	−0.3207, −0.1515	−0.3207, −0.1509
Pvt Inv.				
Mean	−0.6857	−0.5282	−0.4942	−0.4941
90% Interval	−1.0486, −0.3650	−0.8023, −0.2824	−0.7600, - 0.2616	−0.7600, −0.2616
Emp. Income				
Poor Households				
Mean	−1.0235	−0.9482	−0.8879	−0.8869
90% Interval	−1.2329, −0.8106	−1.1403, −0.7542	−1.0683, −0.7050	−1.0657, −0.7050
Rich Households				
Mean	−0.1565	−0.4336	−0.5108	−0.5106
90% Interval	−0.2714, −0.0634	−0.6729, −0.2317	−0.7785, −0.2818	−0.7785, −0.2806

Table 14
Posterior Mean Multipliers ($\theta = 0.10$).

Panel A: Government Schooling Investment				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	0.0621	0.0358	0.2503	0.4932
Pvt Con.	−0.2585	−0.4442	−0.2837	−0.0592
Pvt Inv.	−0.6622	−0.4392	−0.3443	−0.2974
Emp. Income Poor Households.	0.0110	0.3770	0.7476	1.0598
Emp. Income Rich Households.	0.0079	−0.2853	−0.2648	−0.1662
Panel B: Tuition Subsidies Targeting Poor Households				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	−0.7361	−0.6242	−0.3186	−0.0079
Pvt Con.	−0.3527	−0.3172	−0.0093	0.3248
Pvt Inv.	−0.6010	−0.4863	−0.4662	−0.4753
Emp. Income Poor Households.	−2.7194	−1.2813	0.3232	1.7136
Emp. Income Rich Households.	−0.3550	−0.4975	−0.4070	−0.2665
Panel C: Tuition Subsidies Targeting Rich Households				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	−0.6460	−0.6166	−0.3265	−0.0091
Pvt Con.	−0.2427	−0.3999	−0.1932	0.0834
Pvt Inv.	−0.6272	−0.3609	−0.2178	−0.1324
Emp. Income Poor Households.	−0.0382	0.0612	0.1417	0.2041
Emp. Income Rich Households.	0.0032	−0.2826	−0.2452	−0.1220
Panel D: Untargeted Tuition Subsidies				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	−0.5214	−0.4862	−0.2546	−0.0040
Pvt Con.	−0.2051	−0.3051	−0.1283	0.0983
Pvt Inv.	−0.4921	−0.2986	−0.2017	−0.1466
Emp. Income Poor Households.	−0.3470	−0.1183	0.1328	0.3471
Emp. Income Rich Households.	−0.0428 −0.2487	−0.2119	−0.1119	
Panel E: Lump-Sum Transfer to Poor Households				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	−0.0888	−0.1799	−0.2036	−0.2035
Pvt Con.	0.0795	−0.0561	−0.0909	−0.0908
Pvt Inv.	−0.2477	−0.1918	−0.1778	−0.1778
Emp. Income Poor Households.	−0.4492	−0.4324	−0.4189	−0.4185
Emp. Income Rich Households.	−0.0664	−0.1948	−0.2295	−0.2294

Then,

$$\Omega^i_j = 1 - \exp[(1-\beta_j)(Wel^i_{jt} - Wel_j)] \text{ for } j = R \ \& \ P. \tag{5.8}$$

To derive the aggregate welfare cost–benefit of policies, since rich and poor households' consumption differs, I sum the changes in the consumption of both types of households implied by Ω^i_js as follows:

$$\Delta C^i = \omega \Omega^i_R c_R + (1-\omega)\Omega^i_P c_P \tag{5.9}$$

where the aggregate welfare cost–benefit of policy change i, Ω^i, for policy i is expressed as the ratio of the sum of changes in households' consumption and the weighted average of steady-state consumption as follows:

$$\Omega^i = \frac{\Delta C^i}{\omega c_R + (1-\omega)c_P} \tag{5.10}$$

for policy i. If Ω^i_j is positive (negative) a welfare loss (gain) occurs for the jth type of households under policy regime i. Similarly, if Ω^i is positive (negative) an aggregate welfare loss (gain) occurs under policy

Table 15
Posterior Mean Multipliers ($\theta = 0.20$).

Panel A: Government Schooling Investment				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	0.0068	0.1463	0.4780	0.7264
Pvt Con.	−0.2299	−0.2506	0.0218	0.2519
Pvt Inv.	−0.6234	−0.3794	−0.2716	−0.2292
Emp. Income Poor Households.	−0.0455	0.6522	1.3010	1.7017
Emp. Income Rich Households.	0.0028	−0.2043	−0.1160	−0.0019
Panel B: Tuition Subsidies Targeting Poor Households				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	−0.6750	−0.5048	−0.2120	−0.0007
Pvt Con.	−0.3333	−0.2271	0.0614	0.2819
Pvt Inv.	−0.5322	−0.4336	−0.4151	−0.4193
Emp. Income Poor Households.	−2.6845	−1.1022	0.4598	1.4432
Emp. Income Rich Households.	−0.3391	−0.4286	−0.3375	−0.2427
Panel C: Tuition Subsidies Targeting Rich Households				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	−0.5203	−0.4273	−0.1386	0.0867
Pvt Con.	−0.1674	−0.2511	−0.0434	0.1505
Pvt Inv.	−0.5478	−0.2840	−0.1484	−0.0891
Emp. Income Poor Households.	−0.0808	0.0169	0.0891	0.1300
Emp. Income Rich Households.	−0.0116	−0.2338	−0.1695	−0.0675
Panel D: Untargeted Tuition Subsidies				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	−0.4371	−0.3493	−0.1205	0.0552
Pvt Con.	−0.1605	−0.1937	−0.0151	0.1432
Pvt Inv.	−0.4294	−0.2488	−0.1632	−0.1281
Emp. Income Poor Households.	−0.4892	−0.1772	0.1314	0.3244
Emp. Income Rich Households.	−0.0672	−0.2172	−0.1615	−0.0820
Panel E: Lump-Sum Transfers to Poor Households				
Multiplier	Impact	10-Year	25-Year	Long-Term
GDP	−0.0877	−0.1674	−0.1878	−0.1877
Pvt Con.	0.0675	−0.0500	−0.0794	−0.0793
Pvt Inv.	−0.2341	−0.1827	−0.1709	−0.1708
Emp. Income Poor Households.	−0.4566	−0.4255	−0.3907	−0.3903
Emp. Income Rich Households.	−0.0649	−0.1828	−0.2148	−0.2147

regime i.

As welfare costs and benefits depend on the size of government expenditure change, to compare the relative effects of different policy changes, I assume that government expenditure changes by the same amount in the initial period under each policy regime as follows:

$$\Delta gh = \Delta\omega t_{sR} s_R = \Delta(1-\omega) t_{sP} s_P = \Delta(1-\omega) gp = \Delta(\omega s_R + (1-\omega) s_P) t_s. \quad (5.11)$$

To calculate the consumption equivalent, I assume increased government expenditure is equivalent to a 1% rise in public schooling investment from the steady-state ($\Delta gh = .01gh$) to simulate selected models. I report the consumption equivalent in percentage ($\Omega 100$). As in the case of expenditure multipliers, I present mean consumption equivalents and 90% confidence intervals for the model with $\theta = 0.15$ (Table 16), and I only present the mean consumption equivalents for models with $\theta = 0.10$ & 0.20 (Table 17).

The results reveal that poor households' welfare increases from a rise in government schooling investment, targeted or untargeted tuition subsidies and lump-sum transfers (Table 16). In addition, the poor households' welfare gains are statistically significant. In contrast, rich households experience significant welfare loss, with the exception of the tuition subsidy targeting them; however, their welfare gain/loss is statistically insignificant in this case.

Poor households' welfare gain is the highest when tuition subsidy targets them and is the least when it targets rich households. In consumption equivalent terms, poor households' mean welfare gain from the targeted tuition subsidy is 0.024% of their steady-state consumption. The mean welfare gain to poor households from an increase in government schooling investment and untargeted tuition subsidy is much smaller at 0.005%. The lump-sum transfer to poor households also significantly raises their welfare, with the mean welfare gain of 0.017%. As noted previously, an increase in lump-sum transfer significantly raises poor households' consumption and reduces the labour supply, and the amount of leisure increases, despite a rise in time spent schooling. Interestingly, although poor households gain more from tuition subsidies targeting them compared with the lump-sum income transfer.

Rich households' welfare loss from poor households' tuition subsidies and income transfer is the highest. In consumption equivalent terms, the mean welfare loss to rich households from these policies is about 0.0045% of their steady-state consumption. The mean estimated welfare losses from an increase in government schooling investment and untargeted tuition subsidy are 0.0016% and 0.0008%, respectively. The main reason that rich households suffer welfare loss in the case of a rise in government schooling investment and untargeted tuition subsidies and an insignificant gain from tuition subsidies targeting them is that enhanced government expenditure increases taxes. A rise in lump-sum taxes paid by rich households has a positive effect on their labour supply and a negative effect on their consumption and their subsequent welfare.

Poor households' welfare gains from targeted and untargeted tuition subsidies more than offset rich households' welfare loss, resulting in an aggregate welfare rise, particularly in the case of tuition subsidies targeting poor households. The mean aggregate welfare gains from tuition subsidies targeting poor households and untargeted tuition subsidies are respectively estimated to be 0.0031% and 0.0009% of households' steady-state average consumption. For other policies, poor households'

Table 16
Welfare Costs and Benefits in % ($\theta = 0.15$).

Panel A: Government Schooling Investment			
	Poor Households	Rich Households	All Households
Mean	−0.0047	0.0016	0.0005
90% Interval	−0.0080, −0.0020	0.0002, 0.0029	−0.0007, 0.0017
Panel B: Tuition Subsidy to Poor Households			
	Poor Households	Rich Households	All Households
Mean	−0.0238	0.0045	−0.0031
90% Interval	−0.0364, −0.0140	0.0021, 0.0075	−0.0062, −0.0005
Panel C: Tuition Subsidy to Rich Households			
	Poor Households	Rich Households	All Households
Mean	−0.0008	−0.0001	−0.0003
90% Interval	−0.0012, −0.0005	−0.0004, 0.0003	−0.0006, 0.0000
Panel D: Untargeted Tuition Subsidy to Households			
	Poor Households	Rich Households	All Households
Mean	−0.0053	0.0008	−0.0009
90% Interval	−0.0078, −0.0033	0.0002, 0.0016	−0.0018, −0.0002
Panel E: Lump-Sum Transfer to Poor Households			
	Poor Households	Rich Households	All Households
Mean	−0.0169	0.0047	−0.0010
90% Interval	−0.0276, −0.0090	0.0021, 0.0081	−0.0029, 0.0007

Note: Welfare costs and benefits refer to consumption equivalent, where + indicates welfare loss and - implies welfare gain.

Table 17
Mean Welfare Costs and Benefits in % ($\theta = 0.10$ & 0.20).

Panel A: Government Schooling Investment			
θ	Poor Households	Rich Households	All Households
0.10	−0.0021	0.0027	0.0013
0.20	−0.0032	0.0020	0.0009
Panel B: Tuition Subsidy to Poor Households			
θ	Poor Households	Rich Households	All Households
0.10	−0.0254	0.0047	−0.0029
0.20	−0.0231	0.0039	−0.0035
Panel C: Tuition Subsidy to Rich Households			
θ	Poor Households	Rich Households	All Households
0.10	−0.0014	0.0005	−0.0002
0.20	−0.0010	0.0001	−0.0002
Panel D: Untargeted Tuition Subsidy to Households			
θ	Poor Households	Rich Households	All Households
0.10	−0.0057	0.0011	−0.0006
0.20	−0.0054	0.0008	−0.0008
Panel E: Income Transfer to Poor Households			
θ	Poor Households	Rich Households	All Households
0.10	−0.0179	0.0046	−0.0010
0.20	−0.0157	0.0039	−0.0014

Note: Welfare costs and benefits refer to consumption equivalent, where + implies welfare loss and - implies welfare gain.

welfare gain and rich households' welfare loss offset one another, resulting in insignificant gain/loss in aggregate welfare. The pattern and size of welfare gains/losses for poor and rich households and all households are similar for models with $\theta = 0.10$ & 0.20 (Table 17). Overall, these results demonstrate that among all the policies analysed, targeting tuition subsidies to poor households is the superior policy from the perspective of raising poor households' employment income and welfare and aggregate welfare.

6. Conclusion

This study analyses the income and welfare effects of government schooling investment, tuition subsidies and income transfer to poor households. A two-sector DSGE model with human capital is constructed and tested using data from India for the analysis. In the model, schools differ in quality and poor households can only access low-cost and low-quality schools.

The results reveal a considerable heterogeneity regarding these policies' income and welfare effects. In addition, the effects of tuition subsidies depend on whether they are targeted and the type of recipient. The tuition subsidy targeting poor households has the largest positive effect on their human capital, employment income and welfare. The untargeted tuition subsidy also has a significant positive effect on poor households' human capital, employment income and welfare; however, its effect is much smaller compared with the targeted tuition subsidy.

Public schooling investment and lump-sum transfer to poor households have a significant positive effect on their human capital and welfare; however, public investment significantly raises poor households' employment income, while lump-sum transfer significantly reduces it. Results also show that all the spending policies analysed in this paper except for the tuition subsidy targeted at rich have a negative effect on their welfare. Despite their welfare loss, the aggregate welfare rises significantly both in the case of tuition subsidy targeted at the poor and untargeted tuition subsidy.

I conclude with the discussion of some topics for future research. Public policies' income and welfare effects depend on the economic structure. It will be interesting to analyse and quantify the income and welfare effects of different types of public spending on schooling in other developing countries. Public investment in infrastructure is considered to be an important driver of development. It will also be interesting to compare the relative desirability of public spending on schooling and infrastructure in the DSGE framework.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

I have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

I thank the editor and two anonymous referees for their numerous incisive comments that have significantly improved the paper. The responsibility for any errors remaining is entirely mine.

Appendix. Equilibrium conditions

Households

$$1 = \chi((1+tc)p_{ft})^{1-\tau} + (1-\chi)p_{nt}^{1-\tau}; \tag{A.1}$$

$$c_{jft} = \chi c_{jt}((1+tc)p_{ft})^{-\tau} \ \& \tag{A.2}$$

$$c_{jnt} = (1-\chi)c_{jt}p_{nt}^{-\tau} \tag{A.3}$$

for $j = R \ \& \ P$.

Rich Households:

$$\frac{\gamma_t}{c_{Rt}} = \lambda_{ct}^R; \tag{A.4}$$

$$\phi_{lf}\mu_R\gamma_t(l_{Rt} + \phi_{Rs}s_{Rt})^{\zeta} = (1-tw)w_{ft}h_{Rt}\lambda_{ct}^R; \tag{A.5}$$

$$\mu_R\gamma_t(l_{Rt} + \phi_{Rs}s_{Rt})^{\zeta} = MPEL_{Rt}h_{Rt}\lambda_{ct}^R; \tag{A.6}$$

$$MPEL_{Rt} = \alpha_n p_{nt}\frac{y_{nt}}{el_{Rnt}}; \tag{A.7}$$

$$(pf_t v_{Rt} + \phi_{Rs}MPEL_{Rt}h_{Rt})\lambda_{ct}^R = \theta A h_t {s_{Rt}}^{\theta-1}{g_{ht}}^{\theta_g}\lambda_{ht}^R; \tag{A.8}$$

$$\lambda_{ct}^R p_{ft}(1+tc) = \beta_R E_t \lambda_{ct+1}^R[(1-tk)rk_{ft+1} + (1+tc)pf_{t+1}(1-\delta_k)]; \tag{A.9}$$

$$\lambda_{ct}^R = \beta_R E_t \lambda_{ct+1}^R r_{t+1}; \tag{A.10}$$

$$\lambda_{ht}^R = \beta_R E_t[((1-tw)w_{ft+1}l_{ft+1}^R + MPEL_{Rt+1}l_{Rnt+1})\lambda_{ct+1}^R + (1-\delta_h)\lambda_{ht+1}^R]; \tag{A.11}$$

$$y_{nt} = \kappa_n A_t {el_{Rnt}}^{\alpha_n} {el_{Pnt}}^{1-\alpha_n}; \tag{A.12}$$

$$w_{nt} = (1-\alpha_n)p_{nt}\frac{y_{nt}}{el_{Pnt}}; \tag{A.13}$$

$$k_{Rt+1} = (1-\delta_k)k_{Rt} + I_t; \tag{A.14}$$

$$h_{Rt+1} = (1-\delta_h)h_{Rt} + A_{hRt}{s_{Rt}}^{\theta}{g_{ht}}^{\theta_g}; \tag{A.15}$$

$$l_{Rt} = \phi_{lf}l_{Rft} + l_{Rnt} \ \& \tag{A.16}$$

$$ein_{Rt} = (1-tw)w_{ft}h_{Rt}l_{Rft} + [p_{nt}y_{nt} - el_{Pnt}w_{nt}]. \tag{A.17}$$

Poor Households:

$$\frac{\gamma_t}{c_{Pt}} = \lambda_{ct}^P; \tag{A.18}$$

$$\mu_P\gamma_t(l_{pt} + \phi_{Ps}s_{pt})^{\zeta} = w_{nt}h_{Pt}\lambda_{ct}^P; \tag{A.19}$$

$$(pf_t vp_t + \phi_{Ps}w_{nt}h_{Pt})\lambda_{ct}^P = \kappa_h\theta A h_{pt}{s_{Pt}}^{\theta-1}{g_{ht}}^{\theta_g}\lambda_{ht}^P; \tag{A.20}$$

$$\lambda_{ht}^P = \beta_P E_t[\lambda_{ct+1}^P w_{nt+1}l_{Pt+1} + (1-\delta_h)\lambda_{ht+1}^P]; \tag{A.21}$$

$$h_{Pt+1} = (1-\delta_h)h_{Pt} + \kappa_h A_{Pht}{s_{Pt}}^{\theta}{g_{ht}}^{\theta_g} \ \& \tag{A.22}$$

$$w_{nt}h_{Pt}l_{Pt} + pf_t gp_P t = c_{Pt} + pf_t v_{Pt}s_{Pt}. \tag{A.23}$$

Formal Sector Production:

$$y_{ft} = A_t k_{ft}^{\alpha_f}{el_{ft}}^{1-\alpha_f}; \tag{A.24}$$

$$w_{ft} + \psi\left(\frac{el_{ft}}{el_f} - 1\right) = (1-\alpha_f)p_{ft}\frac{y_{ft}}{el_{ft}} \ \& \tag{A.25}$$

$$r_{fkt} = \alpha_f p_{ft}\frac{y_{ft}}{k_{ft}}. \tag{A.26}$$

Market Clearing and Resource Constraints:

$$el_{ft} = \omega h_{Rt}l_{Rft}; \tag{A.27}$$

$$\omega el_{Pnt} = (1-\omega)h_{Pt}l_{Pt}; \tag{A.28}$$

$$el_{Rnt} = h_{Rt}l_{Rnt}; \tag{A.29}$$

$$y_{nt} = \omega c_{Rnt} + (1-\omega)c_{Pnt}; \tag{A.30}$$

$$y_{ft} = \omega c_{Rft} + (1-\omega)c_{Pft} + \omega I_t + g_{ht} + g_{ct} + \omega v_{Rt}s_{Rt} + (1-\omega)v_{Pt}s_{Pt}; \tag{A.31}$$

$$\omega k_{Rft} = k_{ft} \ \& \tag{A.32}$$

$$gdp_t = p_{nt}y_{nt} + p_{ft}y_{ft} - \omega pf_t v_{Rt}s_{Rt} - (1-\omega)pf_t v_{Pt}s_{Pt} - + \frac{\psi}{2}\left(\frac{el_{ft}}{el_f} - 1\right)^2 el_f. \tag{A.33}$$

$$pc_t = p_{ft}(\omega c_{Rft} + (1-\omega)c_{Pft}) + p_{nt}y_{nt}. \tag{A.34}$$

Welfare:

$$Wel_{jt}^i = \ln c_{jt} - \mu_j\frac{(l_{jt} + \phi_{js}s_{jt})^{1+\zeta}}{1+\zeta} + \beta_j Wel_{jt+1} \text{ for } j = R \ \& \ P \ \& \tag{A.35}$$

$$Wel_t^i = \omega Wel_{Rt}^i + (1-\omega)Wel_{Pt}^i. \tag{A.36}$$

Stochastic Processes:

$$\ln\left(\frac{g_{ht}}{g_h}\right) = \rho_{gh}\ln\left(\frac{g_{ht-1}}{g_h}\right) + e_{ght}; \tag{A.37}$$

$$\ln\left(\frac{g_{pt}}{g_P}\right) = \rho_{gP}\ln\left(\frac{g_{Pt-1}}{g_P}\right) + e_{gPt}; \tag{A.38}$$

$$\ln\left(\frac{g_{ct}}{g_c}\right) = \rho_{gc}\ln\left(\frac{g_{ct-1}}{g_c}\right) + e_{gct}; \tag{A.39}$$

$$\ln A_t = \rho_A \ln A_{t-1} + e_{At}; \tag{A.40}$$

$$\ln\left(\frac{A_{jht}}{A_{jh}}\right) = \rho_{Ah}\ln\left(\frac{A_{jht-1}}{A_{jh}}\right) + e_{jAht}\ \&\ \forall j = R\ \&\ P; \tag{A.41}$$

$$\ln\left(\frac{v_{jt}}{v_j}\right) = \rho_{vj}\ln\left(\frac{v_{jt-1}}{v_j}\right) + e_{jv}\ \forall j = R\ \&\ P\ \& \tag{A.42}$$

$$\gamma_t = \rho_\gamma \gamma_{t-1} + e_{\gamma_t}. \tag{A.43}$$

Data availability

Data and replication codes are available at following data repository : https://data.mendeley.com/datasets/6t6tjsk66d/1.

References

Azam, M., Blom, A., 2008. Progress in participation in tertiary education in India from 1983–2004. In: The World Bank Policy Research Working Paper WPS4793. The World Bank.

Baland, J.M., Robinson, J.A., 2000. Is child labor inefficient? J. Polit. Econ. 108, 663–679.

Barro, R., 1991. Economic growth in a cross section of countries. Q. J. Econ. 106 (2), 407–443.

Benhabib, J., Spiegel, M., 1994. The role of human capital in economic development: Evidence from aggregate cross-country data. J. Monetary Econ. 34, 143–173.

De Grip, A., 2006. Evaluating human capital obsolescence. Researchcentrum voor Onderwijs en Arbeidsmarkt Faculteit der Economische Wetenschappen. ROA Working Papers No. 2E available at https://doi.org/10.26481/umarow.200602E.

De Grip, A., van Loo, J., 2002. The economics of skills obsolescence: A review. In: de Grip, A., van Loo, J., Mayhew, K. (Eds.), The Economics of Skills Obsolescence. In: Research in Labor Economics, vol. 21, JAI Press, pp. 1–26.

DeJong, D., Ingram, B., 2001. The cyclical behavior of skill acquisition. Rev. Econ. Dyn. 4 (3), 536–561.

Dissou, Y., Didic, T., Yakautsava, S., 2016. Government spending on education. Hum. Cap. Accumul. Growth, Econ. Model. 58, 9–21.

Duraisamy, P., Duraisamy, M., 2016. Contemporary issues in Indian higher education: Privatization, public and household expenditures and student loan. High. Educ. Futur. 3, 144–163.

Fernández, A., Meza, F., 2015. Informal employment and business cycles in emerging economies: The case of Mexico. Rev. Econ. Dyn. 18, 381–405.

Gabriel, V., Levine, B., Yang, P., 2016. An estimated DSGE open economy model of the Indian economy with financial frictions. In: Ghate C, C., Kletzer, K. (Eds.), Monetary Policy in India a Modern Macroeconomic Perspective. Springer.

Ghate, C., Gopalkrishnan, S., Tarafdar, P., 2016. Fiscal policy in an emerging market business cycle model. J. Econ. Asymmetries 14, 520–577.

Ghate, C., Pandey, I., Patniak, R., 2013. Has India emerged? Business cycle stylized facts from a transitioning economy. Struct. Change Econ. Dyn. 24, 157–172.

GOI, 2012. Report of the Committee on Unorganized Sector Statistics. National Statistical Commision, Government of India, available at https://mospi.gov.in/sites/default/files/committee_reports/nsc_report_un_sec_14mar12.pdf.

Gong, X., van Soest, A., 2002. Wage differentials and mobility in the urban labour market: A panel data analysis for Mexico. Labour Econ. 9, 513–529.

ILO, 2018. Women and Men in the Informal Economy: A Statistical Picture. Technical Report, ILO.

Jacoby, H., Skoufias, E., 1997. Risk, financial markets, and human capital in a developing country. Rev. Econ. Stud. 64 (3), 311–335.

Kass, R., Raftery, A., 1995. Bayes factors. J. Amer. Statist. Assoc. 90, 773–795.

Kuepie, M., C. Nordman, C., Roubaud, F., 2009. Education and earnings in Urban West Africa. J. Comp. Econ. 37, 491–515.

Kumar, A., 2013. Preference based vs. Market based discrimination: Implications for gender differentials in child labor and schooling. J. Dev. Econ. 105, 64–68.

Kumar, A., 2023. Financial market imperfections, informality and government spending multipliers. J. Dev. Econ. 163, 103103.

Kumar, M., Pandey, S., 2021. Wage gap between formal and informal regular workers in India: Evidence from the national sample survey. Glob. J. Emerg. Mark. Econ. 13, 104–121.

La Porta, R., Shleifer, A., 2014. Informality and development. J. Econ. Perspect. 28, 109–126.

Loayza, N., 2016. Informality in the process of development and growth. World Econ. 39, 1856–1916.

Lucas, R., 1988. On the mechanics of economic development. J. Monetary Econ. 22, 3–42.

Mankiw, G., Romer, D., Weil, D., 1992. A contribution to the empirics of economic growth. Q. J. Econ. 107 (2), 407–437.

Meier, S., Sprenger, C., 2010. Present-biased preferences and credit card borrowing. Am. Econ. J. Appl. Econ. 2 (1), 193–210.

Motkuri, V., Revathi, E., 2023. Private expenditure on education in India: National level analysis exploring NSO survey (CES and SCE) estimates. Indian J. Hum. Dev. 17, 131–147.

NCEUS, 2007. Report on Conditions of Work and Promotion of Livelihoods in the Unorganized Sector. Government of India, available at https://www.im4change.org/nceus_reports/NCEUS-2007-Report-on-conditions-of-work-and-promotion-of-livelihoods-in-the-unorganised-sector.pdf.

Nelson, R., Phelps, E., 1966. Investment in humans, technological diffusion, and economic growth. Amer. Econ. Rev. 56 (1/2), 69–75.

Orazem, P., King, E., 2007. Schooling in developing countries: The roles of supply, demand, and government policy. In: Schultz, T., Strauss, J. (Eds.), Handbook of Development Economics, Vol. 4. North Holland, pp. 3475–3559, ch. 55.

Perli, R., Sakellaris, P., 1998. Human capital formation and business cycle persistence. J. Monetary Econ. 42 (1), 67–92.

Prakash, V., 2007. Trends in growth and financing of higher education in India. Econ. Political Wkly. 42, 3249–3258.

Romer, P., 1986. Increasing returns and long-run growth. J. Polit. Econ. 94 (5), 1002–1037.

Romer, P., 1990. Endogenous technological change. J. Polit. Econ. 98 (5), S71–102.

Rossi, F., 2020. Human capital and macroeconomic development: A review of the evidence. World Bank Res. Obs. 35, 227–262.

Schmit-Grohé, S., Uribe, M., 2007. Optimal simple and implementable monetary and fiscal rules. J. Monetary Econ. 54, 1702–1725.

Sengupta, A., Kannan, G., Raveendran, K., 2008. India's common people: Who are they, how many are they and how do they live? Econ. Political Wkly. 43, 49–63.

Sianesi, B., Reenen, J., 2003. The returns to education: Macroeconomics. J. Econ. Surv. 17, 157–200.

Tanaka, T., Camerer, Q., Nguyen, C., 2010. Risk and time preferences: Linking experimental and household survey data from Vietnam. Amer. Econ. Rev. 100, 557–571.

Tilak, J.B., 2009. Household Expenditure on Education and Implications for Redefining the Poverty Line in India, Background Paper, Prepared for the Expert Group on the Review of the Methodology for Estimation of Poverty. Planning Commission, available at chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/http://117.239.78.102:8080/jspui/bitstream/123456789/314/1/Household%20Expenditure%20on%20Education%20and%20Implications%20For%20redi.pdf.

Tilak, J., Chaudhary, P., 2021. Inequality in access to higher education in India between the poor and the rich: Evidence from NSSO data. available at chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/http://www.ecineq.org/ecineq_paris19/papers_EcineqPSE/paper_178.pdf.

UNESCO Report, 2008. Poverty and education. available at https://unesdoc.unesco.org/ark:/48223/pf0000181754.

Unni, J., 2023. Private investment in education and linkage to future employment in india: will the pandemic take its toll? Indian J. Labour Econ. 66, 267–281.

Warner, J., Pleeter, S., 2001. The personal discount rate: Evidence from military downsizing programs. Amer. Econ. Rev. 9, 33–53.

World Bank, 2022. In: Ohnsorge, F., Yu, S. (Eds.), The Long Shadow of Informality. available at https://www.worldbank.org/en/research/publication/informal-economy.

Dr. Alok Kumar is currently a professor of economics at the University of Victoria, British Columbia, Canada. He received his Ph.D. in economics from the Queen's University, Canada. His research interests are macro-economics, monetary economics and development economics. He has published extensively on monetary theory, government spending multipliers, human capital investment, gender bias and foreign aid in leading economics journals.